

**Emotion, Embodiment, and Aesthetic Appraisal:
The Impact of Interoceptive Abilities and Art Type**

Cabbai, G.¹^{*}, Kühnapfel, C.²^{*}, Fingerhut, J.^{3,4}, Kaltwasser, L.³, Prinz, J.⁵, Pelowski,
M.^{2,6}

¹ University of Sussex, School of Psychology, Brighton, UK

² University of Vienna, Faculty of Psychology, Vienna, Austria

³ Humboldt-Universität zu Berlin, Berlin School of Mind and Brain, Berlin, Germany

⁴ Humboldt-Universität zu Berlin, Department of Philosophy, Berlin, Germany

⁵ City University of New York, Graduate Center, New York, USA

⁶ University of Vienna, Vienna Cognitive Science Hub, Vienna, Austria

 These authors contributed equally to this work.

* Corresponding author: Giulia Cabbai, School of Psychology, University of Sussex, Brighton, UK. Pevensey 1 Building, University of Sussex, Falmer, BN1 9QH, UK, (G.Cabbai@sussex.ac.uk),  <https://orcid.org/0000-0002-2309-0315>.

Corinna Kühnapfel, Department of Cognition, Emotion, and Methods in Psychology, Faculty of Psychology, University of Vienna, Vienna, Austria. Wächtergasse 1/301, 1010 Vienna, Austria, (corinna.kuehnafel@univie.ac.at),  <https://orcid.org/0000-0002-1142-6205>.

Author Note

Giulia Cabbai, School of Psychology, University of Sussex; Corinna Kühnapfel, Department of Cognition, Emotion, and Methods in Psychology, Faculty of Psychology, University of Vienna; Joerg Fingerhut, Department of Philosophy, Berlin School of Mind and

Brain, Humboldt-Universität zu Berlin; Laura Kaltwasser, Institute of Psychology, Berlin School of Mind and Brain, Humboldt-Universität zu Berlin; Jesse Prinz, Department of Philosophy, Graduate Center, City University of New York; Matthew Pelowski, Department of Cognition, Emotion, and Methods in Psychology, Faculty of Psychology, and Vienna Cognitive Science Hub, University of Vienna.

Author contribution: G. Cabbai conceived, designed, conducted, and analyzed Study 1. J. Fingerhut and J. Prinz supervised Study 1. L. Kaltwasser assisted in the analysis of the heart rate data in Study 1. C. Kühnapfel and G. Cabbai conceived Study 2. C. Kühnapfel designed, conducted, and analyzed Study 2. M. Pelowski and J. Fingerhut supervised Study 2. G. Cabbai and C. Kühnapfel wrote the manuscript. M. Pelowski, J. Prinz, L. Kaltwasser, and J. Fingerhut provided comments on the finalized manuscript.

The writing of this paper was supported by a grant to MP and JF from the EU Horizon 2020 TRANSFORMATIONS-17-2019, Societal Challenges and the Arts (870827—ARTIS, Art and Research on Transformations of Individuals of Society). For Study 1, GC, JF, and JP were supported by Einstein Foundation, Humboldt-Universität zu Berlin.

Acknowledgements

We thank Dr. Nicolas Rost for their comments and help with the analysis of heart data in Study 1, Dr. Gernot Gerger for comments on the analysis of Study 1, Guiomar del Culvillo Schröder for helping with the data collection for Study 2, Prof. Jamie Ward for sharing and helping with the analysis of the Vicarious Pain Questionnaire for Study 2, and MacKenzie Trupp for discussion.

Abstract

Emotions are considered important in aesthetic experience. Given that emotions have bodily dimensions, it follows that interoception—the ability to perceive bodily signals accurately—may play a role in experiences of art. This raises the intriguing question of whether variation in interoceptive abilities relates to differences in art experience. We addressed this possibility across two studies that employed objective and subjective measures of interoceptive abilities.

In Study 1, we examined the impact of objective interoceptive accuracy (*IAc*) and bodily changes (heart rate and skin conductance) while viewing art on participants' appraisal and emotional response to abstract and representational artworks. *IAc* was positively related to participants' subjective emotional reactions to art, particularly in response to representational paintings, while no relationship was found for art appraisal. We also observed that heart rate change was higher for representational art. Moreover, for the first time, we found that skin conductance response was positively related to art appraisal and emotional response and did not differ by art type.

In Study 2, we assessed the role of individual differences in self-reported interoception and emotion in participants' experiences of art. As in Study 1, we found that individual differences were mainly related to ratings of emotional response, particularly for representational artworks.

All in all, we propose that an interplay between artwork type, emotions, and interoceptive abilities plays a crucial role in the experience of art.

Keywords: interoception, skin conductance, art experience, emotion, embodiment

Embodied cognition theories suggest that the way we perceive the world is deeply linked to our bodies. Bodies not only provide the basis for the way we perceive, engage, and understand our environment, but especially shape our affective and emotional experiences (Azevedo et al., 2020; Azzalani et al., 2019; Critchley & Harrison, 2013). The James-Lange theory of emotions (James, 1884) first proposed that emotional feeling has its basis in the representation and perception of changes in bodily physiology. On these grounds, embodied cognition and contemporary theories of emotions argue for a central role of physiological changes in the definition of emotional experience (Damasio & Carvalho, 2013; Prinz, 2004; Fingerhut & Prinz, 2018; Fuchs & Koch, 2014; Kaltwasser et al., 2017a; 2017b; 2019). The ability by which we can detect, or are subjectively aware of, such subtle changes in the body's internal milieu (including the workings of the heart, muscles, or viscera) and can report these accurately, is called interoception (Craig, 2002).

Studies have shown that people greatly vary in their interoceptive ability (Critchley et al., 2004), which can be assessed in a variety of ways, characterized by distinct dimensions (Garfinkel et al., 2015). *Interoceptive accuracy (IAc)* refers to objective accuracy in detecting internal bodily sensation and is measured via participants' behavioral performance in a specific task in which they track, detect or discriminate internal bodily sensations (e.g., count their heartbeat). In addition, *interoceptive sensibility (IS)* reflects people's beliefs regarding their interoceptive abilities and is typically assessed by means of self-report questionnaires (Garfinkel et al., 2015).

Interoceptive abilities have been empirically connected to emotions. Individuals who exhibit higher interoceptive abilities—as assessed via aforementioned behavioral tasks or self-report measures—show greater sensitivity to emotional responding (Garfinkel et al., 2014; Pollatos et al., 2007; Wiens et al., 2000), report higher emotional intensity (Barrett et al., 2004; Duschek et al., 2015; Jung et al., 2017; Pollatos & Ferentzi, 2018) and reactivity

(Mikkelsen et al., 2019), enhanced integration of bodily signals in emotional experience (Critchley et al., 2004), and exhibit better emotion regulation (Füstös et al., 2013; Kever et al., 2015; Schuette et al., 2021; Zamariola et al., 2019).

This well-documented link between emotion and individual variation in interoceptive abilities suggests that such interpersonal differences may be a key factor in many aspects of our daily life, particularly in situations where emotion may be salient. One such example regards our engagement with and appraisal of art. Art has long been noted to communicate and evoke emotion and emotion is regarded as one of the most fundamental aspects of art experience (e.g., see Gerger et al., 2018; Nummenmaa & Hari, 2023; Tinio & Specker, 2023; Pelowski et al., 2020). One could even argue that through emotions we bring forth the relevant types of perceptual and cognitive changes that characterize art engagement (Fingerhut & Kühnapfel, *forthc.*). The role of emotion in art experience has also been recognized by the growing amount of literature on neuroaesthetics. Empirical studies in this field have consolidated the notion that aesthetic experiences are characterized by a combination of perceptual, cognitive and emotional processes that all have physiological characteristics (for review, see Chatterjee & Vartanian, 2014; Cupchik, 2016). These findings have been captured in a model of aesthetic processing, which proposes aesthetic emotion and appraisal to be the main outputs (Leder et al., 2004; Leder & Nadal, 2014). Sentimentalist accounts have proposed a more intertwined relationship between these outcomes, namely that embodied emotions do play a foundational role in aesthetic judgment (Prinz, 2007; 2011; 2014; also see Fingerhut & Prinz, 2018; 2020).

While no studies have directly addressed the functional link between interoceptive abilities and art experience yet, there is some evidence that speaks to their relationship.

Neuroimaging studies have shown that brain regions associated with emotion and interoception increase activation during aesthetic experience (see Mastandrea et al., 2019 for

a review). For instance, Cupchik et al. (2009) found that aesthetic engagement with paintings was associated with increased activation of the insula, a brain region known to be involved in interoceptive processing (Critchley et al., 2004; Zaki et al., 2012). Activation of the insula during emotional and aesthetic appraisal of art was also reported in subsequent studies (as shown in meta-analyses by Brown et al., 2011, and Vartanian & Skov, 2014). This suggests that the processing of visceral signals is linked with higher-order emotion processing and thus influences the emotive aspects of aesthetic experience (Chatterjee & Vartanian, 2016; Vartanian, 2018).

Clinical evidence also hints at a link between interoceptive processing and art experience. Altered interoceptive states are hallmarks of alexithymia (characterized by a deficit in the recognition of one's own emotions, see Donges et al., 2014), depression (Eggart et al., 2019), and anxiety (Brewer et al., 2016). A common characteristic of these clinical populations is that they tend to have a low interest or sensitivity toward art (Taylor et al., 1991; Bagby et al., 1994; Giannini et al., 2013). For example, individuals with anhedonia, which is characterized by low mood and decreased ability to experience pleasure, exhibit diminished experiences of beauty (Brielmann & Pelli, 2019)¹. Similarly, people with alexithymia tend to show a “relativistic” approach to art (i.e., they say that beauty is a matter of subjective points of view; Prinz, 2014).

Despite the compelling amount of evidence, this empirical line of study is only at the beginning, as it has so far mostly involved clinical populations or assessments of interoception in terms of activity in associated brain areas, but not yet via behavioral tests of interoceptive abilities. To date, the vast majority of studies that have demonstrated the role of individual differences in interoceptive abilities used standardized sets of emotional stimuli

¹ Indeed, one of the items used in an anhedonia diagnosing questionnaire (Physical Anhedonia Scale; PAS) reads “The beauty of sunsets is greatly overrated” (Chapman et al., 1976).

such as photographs of daily life events or faces (taken for instance from the International Affective Picture System, IAPS; Bradley & Lang, 2007). Importantly, art is often more complex than these emotional stimuli and places different demands on us, such as contemplation and appraisal. Thus, whether individual differences in interoception could also shape art experience has not yet been addressed. Moreover, the functional role and the actual impact of physiological responses on the experience of visual art and how this is modulated by individual differences in interoceptive abilities is yet unclear.

The present project aimed to shed light on the role of interoception in art experience by investigating how bodily responses and individual differences in interoception connect to emotion and appraisal of art. In two studies, we systematically assessed individuals who differed in their interoceptive abilities by using objective and subjective measures, and examining how this trait would affect reports of emotional response and appraisal when viewing art. By exploring how interoception and bodily responses shape art experience, our research sought to further characterize the highly individualized responses to art. Indeed, paintings experienced as moving or as beautiful differ widely across people (Vessel et al., 2012; 2013; Jola et al., 2014; Specker et al., 2020; Gerger et al., 2018; Schlotz et al., 2021; Williams et al., 2018). We therefore aimed to bring empirical support to the idea that aesthetic judgments and emotional reactions do not just depend upon external features, such as properties of the artwork and the emotion depicted, but also on how individuals process emotions subjectively and physiologically.

Study 1

Study 1 investigated the link between interoception and art experience by combining three lines of assessment, namely (1) subjective measures of emotional response and appraisal of artworks, (2) physiological body response (Skin Conductance Response, Heart

Rate change), and (3) individual differences in *IAc* (i.e., the ability to perceive one's internal bodily state, e.g., heartbeat).

First, we predicted stronger bodily responses (indexed by increased Skin Conductance Response and Heart Rate change) and higher subjective emotional response in the case of representational artworks, since these have been found to be more direct in conveying emotions (through faces and body language) than abstract artworks and resemble what may occur in interpersonal or social settings pertaining to daily life situations (Gerger et al., 2018).

Second, we hypothesized that bodily changes would have an effect on the emotional experience and appraisal of artworks. In particular, we predicted a positive relationship between the intensity of the physiological signals and participants' subjective ratings. We also considered whether art type influenced this relationship. If representational art induces a stronger emotional reaction, we expected that the link between physiology and art experience ratings would be stronger for this type of art, compared to abstract art.

We also expected to find a positive relationship between subjective artwork ratings and an individual's *IAc*. This would support that interoception is an essential component of art experience linked to the more emotionally intense enjoyment of art. However, we also considered whether this relationship would be more pronounced for representational than abstract art.

Finally, we assessed whether and how *IAc* moderates the coupling of bodily responses and art experience. We hypothesized that the more accurately participants could track their heartbeat, the stronger the observed link between Heart Rate reactions or sympathetic arousal (indexed by Skin Conductance Response) and the subjective ratings. This would provide evidence that the ability to introspect relates to the tendency to use bodily responses as information for aesthetic judgment (in line with aesthetic sentimentalism, Prinz, 2007; 2011;

2014; Prinz & Fingerhut, 2018; 2020). Finally, we investigated whether the type of artwork (representational vs. abstract) influences this relationship.

Method

Participants

Forty right-handed adults were recruited for the study (20 female, 19 male, $M_{age} = 25.97$, $SD = 3.5$ years). One participant was excluded because this person did not understand the Heartbeat Perception Task (HPT; Schandry, 1981), resulting in a final sample of 39 participants (19 female; $M_{age} = 26.10$; $SD = 3.46$, range 18-35). All participants provided informed consent before taking part and received 9 Euros per hour as compensation. All were naïve to the purpose of the study and declared to not have any psychiatric, neurological, or cardiac conditions. The ethical approval of the study was granted by the Ethics Committee of the Psychology Department of [redacted for peer review] and all the procedures followed the principles of the Declaration of Helsinki (2013).

Stimuli

The stimuli consisted of 30 abstract and 30 representational artworks from the 20th and 21st centuries by 59 different artists (e.g., Willem de Kooning, Paul Klee, Francis Bacon, see Supplementary Material Table S1 for the full list of artworks and artists). All images were selected from online sources, with the aim to allow for a range of emotion and appraisal ratings and with the restriction of having a resolution of at least 1000 pixels. Following Gerger et al. (2018), the distinctive feature characterizing our representational artworks was the presence of human bodies, bodily parts, or portraits. Artworks were classified as representational if they depicted humans in emotional scenes. Abstract artworks were defined by the absence of any recognizable object or human body part. Following Vessel et al. (2012), we did not choose artworks often reproduced in media, to minimize familiarity.

Procedure

The experiment took place in a laboratory room maintained at a constant temperature of approximately 20°C. Participants were seated on a chair throughout the entire duration of the experiment, facing a computer screen (displaying the experimental stimuli) positioned approximately 90 cm away.

The experimental session consisted of three main parts: 1) *IAc* assessment, 2) a resting state period, and 3) the artwork rating task. In the *IAc* assessment, participants underwent a customized version of the HPT (see Interoception Accuracy Assessment section) after electrocardiogram (ECG) electrodes were placed on them. Following that, participants were equipped with electrodes for measuring galvanic skin responses and underwent an 8-minute resting period that served as habituation period. Subsequently, participants completed the artwork rating task using a mouse. At the end of the experimental session, participants filled out a brief demographic survey.

Interoceptive Accuracy Assessment

In terms of individual differences in interoception, in the present study we focused on assessing *IAc*. To date, the most investigated system for assessing interoceptive accuracy has been the cardiovascular one, since compared to other visceral signals, heartbeat is easier to discriminate and measure (Desmedt et al., 2023); thus, experimental evidence mainly relies on heartbeat perception and sensitivity to cardiac awareness. Specifically, *IAc* has been frequently assessed using the Heartbeat Perception Task (HPT; Schandry, 1981). In this task, participants are asked to count their heartbeat in given time windows. Their reported count is then compared with the actual number of heartbeats measured via an electrocardiogram (ECG) to calculate an *IAc* score. Using this task, research has examined the role of interoception in the degree to which our emotions are influenced by bodily signals. For example, Dunn et al. (2010) demonstrated that the more accurately participants could track their heartbeat, the stronger the relationship between their bodily reactions (*HR* change) and

their subjective arousal (but not valence) towards affective stimuli. Christensen et al. (2021) found that higher *IAc* was associated with a stronger relationship between ratings of expressivity of dance movement clips and skin conductance response while watching the same clips.

We implemented a customized version of the HPT in MATLAB (R2014b; The Mathworks Inc., Natick, MA, USA) prior to the artwork rating task. Before the start of the HPT, participants were instructed to sit with both feet on the ground and their hands positioned with palms up on their knees. On each of the 5 trials, participants were instructed to count how many heartbeats they felt over varying time intervals (15, 25, 35, 45, and 55 s) presented in randomized order, delimited by the onset and offset of two different tones. Subsequently, after each count, participants had to indicate on a 5-point scale how confident they felt about their estimate. They were requested not to take their own pulse or to try any other manipulations that might have facilitated the detection of their heartbeats, such as counting seconds. *IAc* was calculated by comparing counted heartbeats with the number of heartbeats recorded by the ECG following Schandry (1981) and McCall et al. (2015). Specifically, we used the following formula: $1/5 * \sum [1 - (|\text{recorded heartbeats} - \text{counted heartbeats}|) / \text{recorded heartbeats}]$ resulting in values ranging from 0 to 1.

Artwork Rating Task

The behavioral artwork rating part of the experiment (comprising the resting state) was created with the Psychopy toolbox running on Python 2.7 (Peirce, 2007). The task was characterized by a within-subject design, involving two main conditions which corresponded to the two types of artworks being presented to participants: abstract and representational. On every trial, participants were presented with an artwork (abstract or representational) for 8 s, preceded by a 5-s inter-trial interval (ITI), consisting of a fixation cross. Immediately after each artwork presentation, we collected two ratings. More specifically, we addressed two

dimensions proposed in psychological models of art perception, according to which art experience generally involves an aesthetic assessment, as well as affective evaluation (Chatterjee & Vartanian 2014; Leder et al., 2004; Leder & Nadal, 2014; Pelowski et al., 2016). Specifically, we chose *Liking* as appraisal rating and *Being Moved* to capture emotional experience. Regarding *Being Moved*, this construct is considered to capture the intensity of the emotional response and engagement with artworks (Menninghaus, 2015; Schindler et al., 2017; Høffding et al., 2022; Vessel et al., 2012; 2013). *Being Moved* represents engagement irrespective of valence or arousal, and it thus can explain the attraction of engaging with negatively valenced emotion, such as sad music (Juslin, 2013). Participants rated *Liking* (“How much do you like this image?”) and *Being Moved* (“How much are you moved by this image?”) on a 7-point Likert-type scale (1, *not at all*; 7, *very much*).

To control for order effects, the order appearance of the two conditions (representational first or abstract first) and the rating scales (*Being Moved* or *Liking* first) was counterbalanced between participants but fixed within each participant’s session. Abstract and representational artworks were presented in separate blocks, for a total of four blocks each composed of 15 trials. Artwork order was randomized within the block. The inter-block intervals (IBI) were 45 seconds. The sequence of blocks and trials is illustrated in Figure 1.

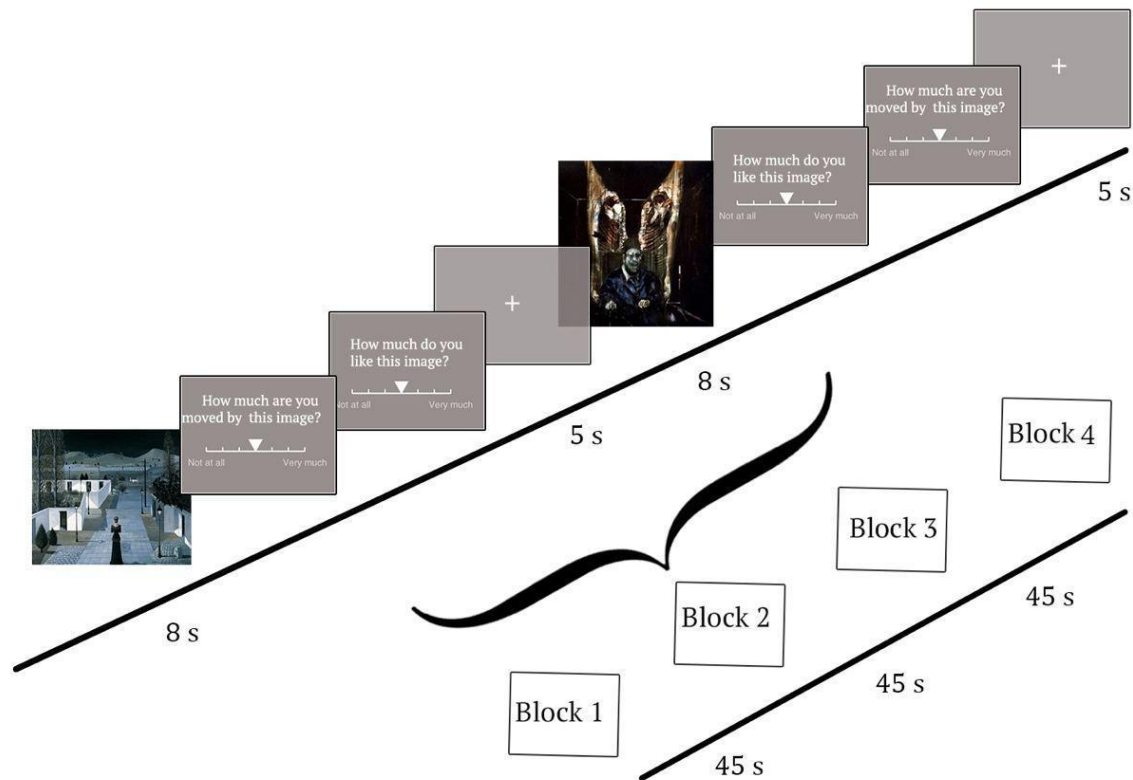


Figure 1. Schematic representation of the main task. On every trial, participants were shown an artwork (8 s) followed by two Likert Scales (*Liking* or *Being Moved*).

Physiological Recordings

At the physiological level, we assessed heart rate (*HR*) change and skin conductance response (*SCR*) while participants viewed artworks as an index of bodily responses to art, which have been implicated in emotional responses (Kreibig, 2010). *HR* is characterized by multiphasic responses, with a deceleration phase prior to a threatening experience (which indicates attentional orienting) usually followed by a consistent increase in its rhythm (Lang et al., 1993). An overall greater *HR* deceleration throughout the rapid and repeated exposition to affective stimuli is considered a general indicator of higher arousal, which we used for the present experiment. Instead, an increase in *SCR* has been linked to intense emotions such as fear, anger, amusement (Kreibig et al., 2010) and more generally, increased arousal (Critchley, 2002).

Heart rate. Heart rate was recorded with an ECG module that was linked to an amplifier (BrainAmp; Brain Products) via its Trigger Box with a sampling rate of 500 Hz. Two electrodes were placed on the left and right lower ribcage respectively and a reference electrode was attached to the sternal end of the right clavicle. According to general recommendations (Fowles et al., 1981), electrodes were always gelled beforehand, and the skin underneath was cleaned by using an alcohol patch.

Skin conductance. Skin conductance was recorded by means of Brain Products' external Galvanic Skin Response (GSR) module. Two pre-gelled Ag/AgCl finger electrodes attached with Velcro straps were placed on the medial phalanges of the left index and ring fingers respectively (of participants' non-dominant hand). The signal was recorded with a sampling rate of 1 kHz.

For details on the physiological data preprocessing and analysis, see Supplementary Material.

Post-Study Survey

We collected additional demographic information via an online survey implemented in *Qualtrics*, which was administered at the end of the experimental session. Specifically, we asked questions concerning education, participant's frequency of museum visits (Never to weekly), preference for abstract or representational art (two choices, abstract or representational), art courses (0 to more than 4), art training (no to more than 2 years), how much they liked art (Likert Scale, 1, *not at all*; 10, *extremely*), as well as how much they considered it important that the ideal body was represented in art (slider ranging from 0 to 100). Last, self-reported political orientation (Likert scale, 1, *very conservative*; 10, *very liberal*).

Statistical Analysis

Statistical analysis was run in *R* (version 4.2.0 - R Core Team, 2018). To test our hypothesis, we ran linear mixed effect models (LMEMs), which provide a flexible approach for the analysis of data from repeated measurements and avoid information loss due to averaging over observations or participants, thus controlling for false positives (Judd et al., 2012). By introducing random effects for every participant and artwork, this approach accounts for variation across participants and trials assigning a different intercept value to each participant and trial. Mixed models were run with the function *lmer()* from the *lme4* package version 1.1-21 (Bates et al., 2014). Significance was calculated using the *lmerTest* package (Kuznetsova et al., 2017), which applies Satterthwaite's method to estimate degrees of freedom and generates p-values for mixed models. The alpha level was set to .05. To further test significant interaction effects, the functions *emtrends* and *emmeans* from the package *emmeans* (Lenth et al., 2023) were used. This study was not preregistered hence results are exploratory.

Results

32 out of the final 39 participants fully completed the post-study survey administered after the experiment. Overall, the participants indicated an above average appreciation of art in general ($M = 7.78$, $SD = 1.83$, $range = 3-10$). When asked about art preference, 59.37% of the responders showed a preference for representational art whereas 40.62% preferred abstract art. More than half of the participants had never undertaken any art course (53.12%) or art training (68.75%, see Supplementary Material Table S2 for full survey results).

Inter-Rater Agreement in Emotional Response and Appraisal

Prior to our individual differences analyses, we checked whether there was variability in artwork assessment across participants. To this end, we computed the intra-class correlation coefficient (ICC) to assess the agreement between our participants in rating the 60 artworks on the dimensions of *Liking* and *Being Moved*, with the ICC function from the *psych*

package (Revelle, 2017) in *R*. Using the two-way random effect models and “single rater” unity, we found “none” level of agreement (McHugh, 2012) between participants regarding how much they liked each painting ($\kappa = 0.11, p < .001$) and how much they were moved by it ($\kappa = 0.19, p < .001$). This reflected that subjective responses were highly individualized (in line with Vessel et al., 2012; 2013).

Effect of Art Type on Art Experience and Bodily Responses

Repeated measures correlations (using *rmcorr* package implemented in *R*, Bakdash & Marusich, 2017) revealed that within-participants ratings of *Liking* and *Being Moved* were positively correlated, $r(37) = 0.47, p < .001, 95\% \text{ CI } [0.44, 0.50]$.

To test for differences in subjective ratings and physiological responses depending on the type of artwork shown (abstract vs. representational), we ran two different mixed models for each subjective rating (*Being Moved*, *Liking*) and for each physiological response (*HR*, *SCR*). Subjective ratings or psychological responses were the outcome variable, while art type was entered as a fixed effect and random intercepts were added for both participants and artwork.

The effect of art type was significant for *Liking*, $F_{1, 58} = 7.08, p = .01$, and *Being Moved*, $F_{1, 58} = 31.83, p < .001$. Representational art was characterized by higher *Liking* ratings ($M = 4.02, SE = 0.16$) than abstract art ($M = 3.60, SE = 0.16$). Similarly, *Being Moved* ratings were higher for representational ($M = 4.30, SE = 0.17$) than abstract artworks ($M = 3.38, SE = 0.17$), see Figure 2.

Concerning physiological responses, we found a main effect of art type on *HR*, $F_{1, 60.51} = 8.11, p = .006$, with greater *HR* deceleration for representational ($M = -0.06, SE = 0.03$) than abstract artworks ($M = 0.06, SE = 0.03$), see Figure 2. We found no effect on *SCR*, $F_{1, 396.47} = 0.76, p = .38$.

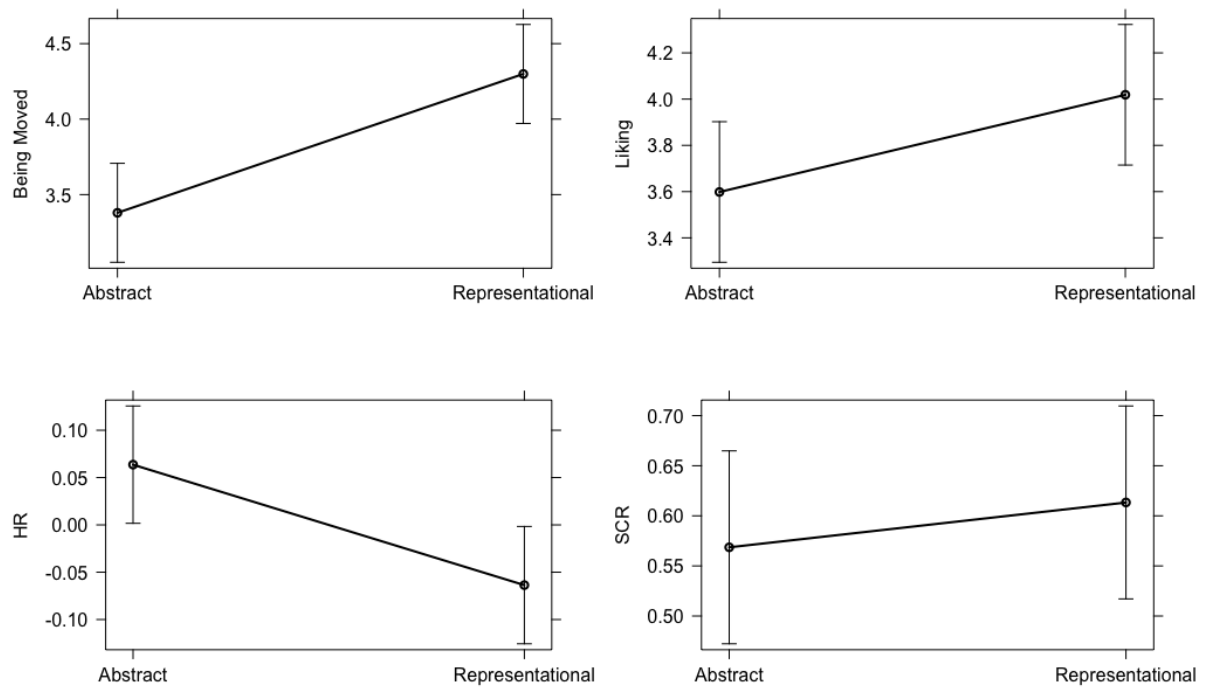


Figure 2. Effect of art type (abstract vs. representational) on *Liking* and *Being Moved* (upper panels) and *HR/SCR* (bottom panels). Only *Being Moved*, *Liking* and *HR* significantly differed between art types.

Effect of Bodily Responses on Emotional Response and Appraisal

To investigate the impact of bodily responses and subjective responses to art, we ran separate additional models with *Liking* or *Being Moved* as the main outcome and *SCR* or *HR* as main effect, while keeping the same random effect structure as our previous models. We note that across all studies presented here, we also standardized continuous predictors prior to running the model using the *scale()* function implemented in R to ease interpretation of the model output (coefficients and standard error).

We observed a significant main effect of *SCR* on *Liking*, $b = 0.06$, $F_{1, 2270.8} = 5.32$, $p = .021$, and *Being Moved*, $b = 0.05$, $F_{1, 2259.2} = 4.85$, $p = .03$, see Figure 3. In contrast, there was no significant main effect of *HR* change on *Liking*, $F_{1, 2259.3} = .19$, $p = .66$, and *Being Moved*, $F_{1, 2251.3} = .02$, $p = .88$.

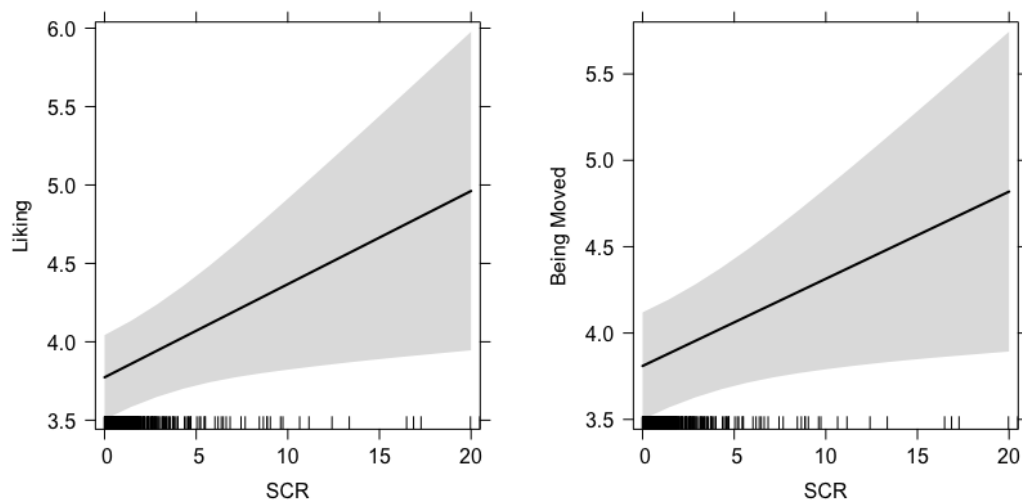


Figure 3. Main effect of *SCR* on *Liking* and *Being Moved*.

To assess whether bodily responses would differentially affect subjective ratings depending on the type of artwork being presented, in a second model we included art type as an interaction term. We found a significant interaction between art type and *HR* in predicting *Being Moved*, $F_{1,2257.87} = 5.53, p = .02$. *HR* deceleration was significantly greater for *Being Moved* ratings when participants viewed representational paintings ($trend = -0.072, SE = 0.04$), compared to when viewing abstract paintings ($trend = 0.058, SE = 0.037$), see Figure 4. There was no significant interaction effect between art type and *HR* on *Liking*, $F_{1,2264.42} = 3.46, p = .06$. Similarly, no significant interaction was found between art type and *SCR* in predicting *Liking*, $F_{1,2265.97} = 3.46, p = .06$, and *Being Moved*, $F_{1,2258.80} = 0.67, p = .41$.

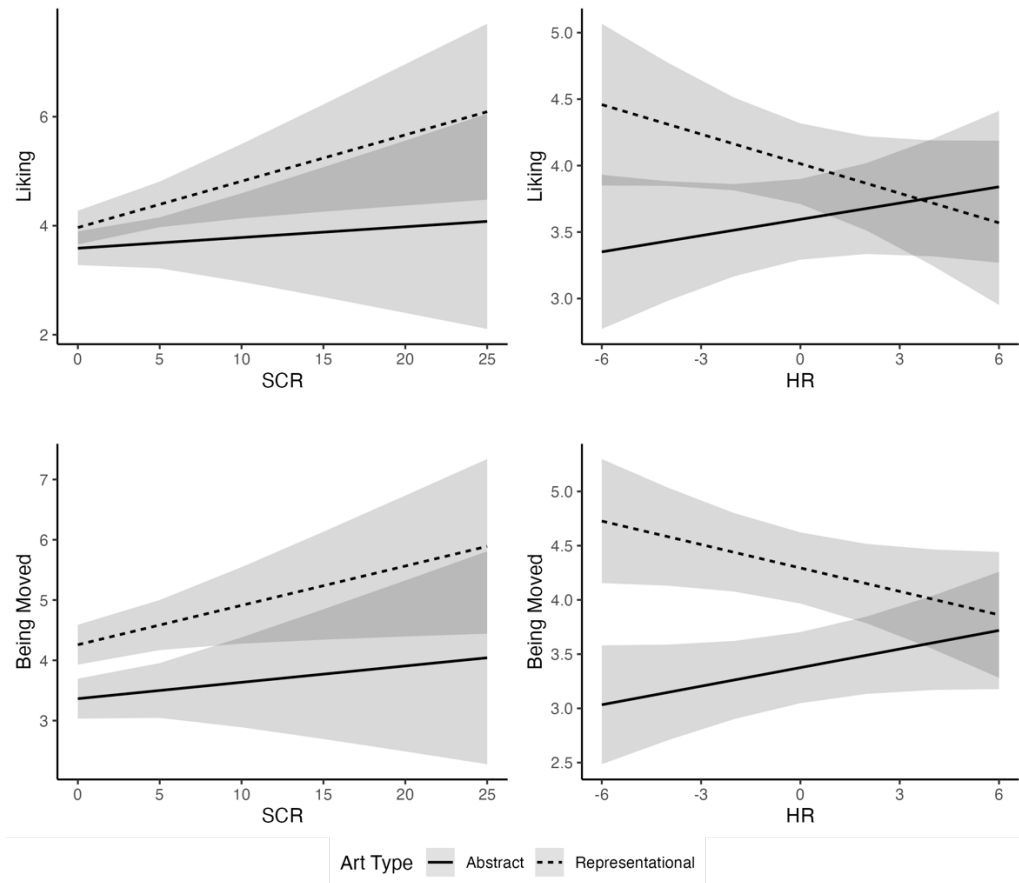


Figure 4. Effect plots depicting the interaction effect between art type (representational, abstract) and *SCR/HR* on *Liking/Being Moved*. *HR* was significantly higher for *Being Moved* ratings of representational compared to abstract artworks. All other effects were n.s.

Effect of Individual Differences in *IAc* on Emotional Response and Appraisal

In our sample, *IAc* scores ranged from 0.50 to 0.96 ($M = 0.72$, $SD = 0.13$), in line with literature using the HPT (e.g., Arslanova et al., 2022; Uraguchi et al., 2022). To test whether participants' *IAc* scores affected art experience, we ran a model with *IAc* as a main predictor, the ratings as outcome variables, and participants and stimuli included as random effects. *IAc* was treated as a continuous variable based on previous guidelines (Ehlers, 1998; Schuette et al., 2021).

Contrary to our expectations, there was no significant effect of *IAc* on *Liking*, $F_{1,37} = 0.06$, $p = .81$, and *Being Moved*, $F_{1,37} = 0.02$, $p = .88$. In a second step, we added the

interaction term with art type to assess whether the effect of *IAc* on the ratings varied as a function of the artwork content. We found a significant interaction effect between *IAc* and art type in predicting *Being Moved*, $F_{1, 2241} = 4.54, p = .033$, and a marginally significant interaction effect in predicting *Liking*, $F_{1, 2241} = 3.38, p = .06$. Opposite trends characterized the impact of *IAc* on *Being Moved* ratings: specifically, higher *IAc* was related to higher *Being Moved* ratings of representational art ($trend = 0.61, SE = 0.95$), suggesting that people with higher *IAc* were more moved than people with lower *IAc* when viewing this type of art, see Figure 5. In contrast, a negative trend was found for abstract art ($trend = -0.31, SE = 0.99$). The positive trend for representational art was almost double the negative trend for abstract art, suggesting that the effect of *IAc* on *Being Moved* ratings was stronger for representational than abstract artworks.

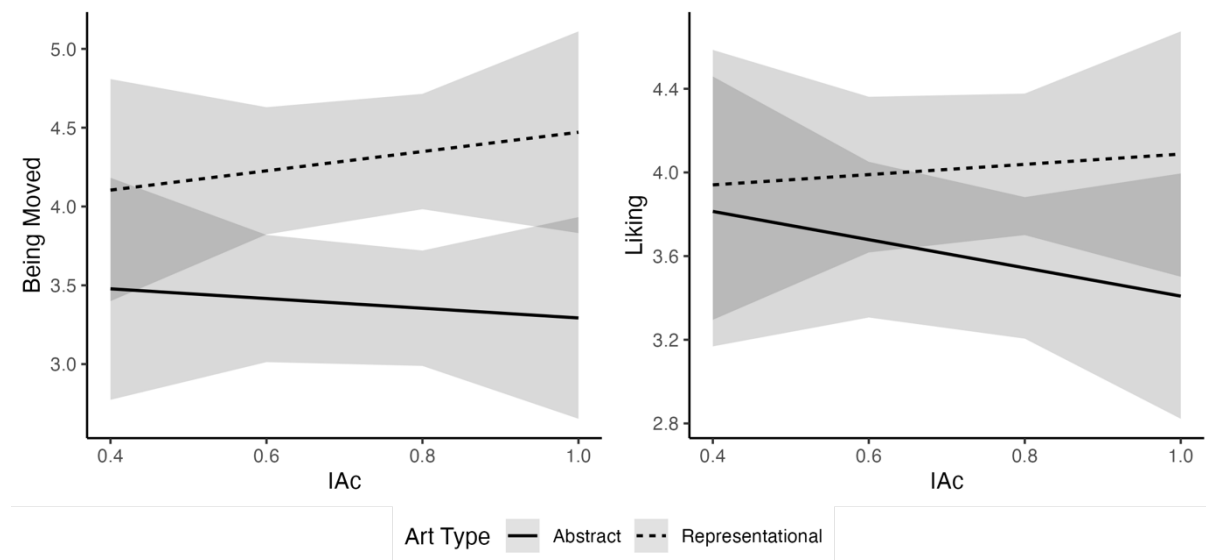


Figure 5. Interaction effect between artwork type (representational vs. abstract) and *IAc* on artwork ratings (*Being Moved*, *Liking*).

Effect of *IAc* as a Moderator on the Relationship between Bodily Responses and Emotional Response/Appraisal

To test whether the relationship between bodily changes and art experience was dependent on how accurately individuals perceive bodily activity, in our model we included

the interaction between *IAc* and *HR/SCR* for each rating, keeping the same random effects structure of previous models.

IAc moderated the coupling between *SCR* and *Liking*, $F_{1, 2279.6} = 5.46$, $p = .02$. Specifically, the relationship between *SCR* and *Liking* was stronger and more positive for individuals with lower *IAc* scores ($trend = 0.26$, $SE = 0.08$) compared to individuals with lower *IAc* scores ($trend = -0.032$, $SE = 0.05$), see Figure 6. *IAc* did not moderate the relationship between *SCR* and *Being Moved*, $F_{1, 2258.9} = 2.13$, $p = .14$. Also, the moderation of *IAc* was not significant for the relationship between *HR* and *Being Moved*, $F_{1, 2250.2} = .88$, $p = .35$. The interaction between *HR* and *IAc* was only marginally significant for *Liking*, $F_{1, 2258.1} = 3.77$, $p = .052$.

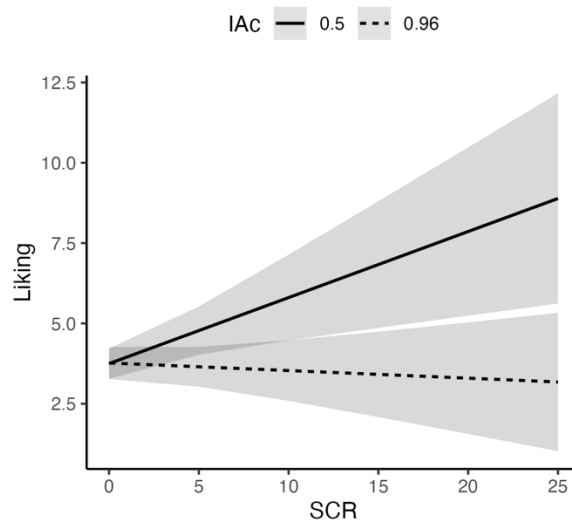


Figure 6. *Liking* rating as a function of *SCR* and *IAc*. All other effects were n.s.

Discussion Study 1

Study 1 results showed a complex interplay between physiological responses, individual variation in interoception, and art experience. In general, although responses to art stimuli were highly individualized across participants, we observed that art appraisal (*Liking*) and emotional engagement (*Being Moved*) were affected by both external information (art stimuli) as well as physiological responses (*HR* and *SCR*) during the art viewing periods.

First, in line with our hypotheses, ratings differed between art types and were both higher for representational than abstract artworks. The higher ratings of *Being Moved* align with previous studies showing a more impactful conveyance of emotions in this type of art, compared to abstract art (Leder et al., 2004; Schepman et al., 2015; Rodway et al., 2016; Gerger et al., 2018). Our behavioral results also indicate a relationship between appraisal and emotion: we found a moderate correlation between *Liking* and *Being Moved*. This is in line with theoretical arguments that emotional engagement (e.g., *Being Moved*) is tightly linked and may contribute to the aesthetic appreciation and evaluation of artworks (Fingerhut & Kühnapfel, *forthc.*; Fingerhut & Prinz, 2018; Prinz, 2011; 2014).

In terms of physiological responses, *HR* deceleration was greater for representational artworks than abstract artworks. These findings suggest that representational art generally elicits stronger subjective appraisal and emotional responses as well as more physiological arousal (indexed by *HR*). In contrast, *SCR* did not vary between art types.

Crucially, here we also show that our subjective measures relate to physiological responses. In particular, we found that *SCR* directly predicted participants' ratings. Specifically, an increase in *SCR* was associated with higher *Liking* and *Being Moved* ratings, and this was not affected by art type. To our knowledge, this is the first time that a direct link between *SCR* and the experience of visual art has been reported. This result is in line with the idea of a relationship between art appraisal and embodied emotions (e.g., Fingerhut & Prinz, 2018), as well as theories relating appraisal to physiological arousal (e.g., see Berlyne, 1971; 1974). In contrast, the effect of *HR* on subjective ratings was less direct than *SCR* and varied as a function of art type: more pronounced *HR* deceleration was associated with higher *Being Moved* ratings for representational art, whereas the opposite pattern was observed for abstract art. Pollatos et al. (2007) previously reported a more pronounced decrease in *HR* for more negatively-valenced images. Although we did not collect valence ratings in the present study,

we note that some of the representational artworks included potentially disturbing content, such as Francis Bacon's "Pope Innocent" (1953) and Franz von Stuck's "Inferno" (1908) that may be appraised as negatively valenced and could have ultimately affected *HR*. Such a possibility could be addressed in future studies by collecting valence ratings of our artwork set; we addressed this in Study 2.

Contrary to our predictions, we found no evidence that ratings of artworks would be directly related to variation in *IAC*. Our results suggest a more nuanced view, whereby *IAC*'s impact on ratings varies as a function of art type. Specifically, higher *Being Moved* ratings were related to higher *IAC* for representational artworks only, whereas a smaller and opposite trend was found for abstract artworks.

Concerning the question of whether *IAC* moderated the coupling between bodily responses and art experience, our results were mixed. We found a significant moderation of *IAC* in the relationship between *Liking* and *SCR* but in the opposite direction of what we expected. Specifically, the coupling between *Liking* and *SCR* was stronger and more positive for low scores in *IAC*, compared to high *IAC* scores. On a first glance, this result appears in contrast to previous literature reporting that people with high *IAC* are characterized by a stronger relationship between physiology and subjective feelings of arousal towards emotional images compared to those with lower *IAC* (e.g., Dunn et al., 2010; Pollatos et al., 2007). We note, however, that these studies investigated only *HR* change, not *SCR*. Since *IAC* measures an individual's accuracy in counting their heartbeats, it may not reflect accuracy with *SCR*. That could account for our failure to find the predicted direction of relationship here. We did find that *IAC* was a marginally significant moderator of the link between *HR* and appraisal ratings. Future work with a larger sample and other cardiovascular measures could further explore this potential relationship.

Additionally, as mentioned in the Introduction, *IAc* is just one measure of interoceptive abilities. It could be that other dimensions of interoception (e.g., interoceptive sensitivity; Garfinkel et al., 2015) would map onto art experience ratings. For instance, research has demonstrated that interoceptive accuracy and sensibility can be dissociated and be independently related to measures of interest (Garfinkel et al., 2015; Schuette et al., 2021). Moreover, while research has found a link between *IAc* and simple affective ratings (e.g., arousal, see Dunn et al., 2010), the ratings we collected involved complex psychological dimensions underlying art experience. More simple affective ratings of valence and arousal to art may be more strongly associated with interoceptive dimensions. As these aspects have never been studied, we explored them in Study 2.

Study 2

Study 2 aimed to extend the results of Study 1 by exploring additional measures of emotional response to art and individual differences in interoception and emotion processing. This employed the same artwork stimuli and rating task as in Study 1. In addition to collecting subjective ratings of *Liking* and *Being Moved*, we also collected more simple affective ratings (i.e., *Valence* and *Arousal*). We also considered a different dimension of interoceptive abilities by employing a subjective self-report measure of interoceptive sensitivity (Garfinkel et al., 2015).

IS, which reflects subjective sensitivity across a range of bodily domains, had never been assessed in relation to art. With regard to variation in emotional experience (Vessel et al., 2012), we included three additional individual difference measures associated with emotional processing: Need for Affect, which is a trait characterized by a desire to feel intense emotional states (assessed via the Need for Affect-short scale; Appel et al., 2012), and the propensity to which an individual feels intense aesthetic and bodily experiences when engaging with art (assessed via the Aesthetic Experiences Scale; Silvia & Nusbaum, 2011).

Individuals high in need for affect have been shown to seek out and become more deeply involved in aesthetic/art interactions (Bartsch et al., 2010; Maio & Esses, 2001) and may also have the propensity to seek out emotional as opposed to more cognitive/conceptual-focused interactions (Appel et al., 2012). Finally, we were also interested in individual differences in vicarious pain perception (as assessed via the Vicarious Pain Questionnaire; Grice-Jackson et al., 2017). Indeed, individuals who experience bodily sensations of pain when viewing another person in pain, so-called vicarious pain (VP) responders, score higher in interoception measures and show brain differences in regions associated with bodily self-awareness (Bowling et al., 2019). Based on this, we investigated whether VP-responders would differ in their emotional appraisal of art, as they also score higher in empathy assessments (e.g., Lockwood et al., 2016; Young et al., 2017), which has been shown to be fundamental to emotional experience (Gerger et al., 2018).

Concerning subjective ratings, we expected to replicate the result from Study 1 that *Liking* and *Being Moved* ratings would be higher for representational artworks. We also tested whether *Valence* and *Arousal* would vary between artwork types. Concerning individual differences in interoception, we expected participants with higher *IS* (assessed via the MAIA-2; Mehling et al., 2018) to be characterized by higher levels of emotional response and appraisal, particularly for representational art. In terms of individual differences in emotion, we tested the relationship between the three individual difference measures (AES, NAQ, VP) and art experience.

Methods

Study 2 was pre-registered on Open Science Framework (OSF) under <https://osf.io/da2pk>.

Participants

The study involved a final sample of 222 participants (154 female, 65 male, 3 other, $M_{age} = 23.02$, $SD = 3.82$, range = 18 to 44, see Table S3 for full demographics). The study followed the guidelines of the ethics committee of the University of [redacted]. Participants received course credit for their participation.

This sample size was based on an a priori power analysis using G*Power (Faul et al., 2007) to detect main effects larger than a medium-sized effect having an f^2 of .25 according to Cohen (1988) with a power of 0.8, which suggested a target of 184. As indicated in the preregistration, we expected around 10% potential data loss for technical reasons and the study running online. In the attempt to complete data collection within the limited time afforded to the data collection (involving the providing of class credits), we recruited a larger number of 224 participants by the time we observed that the stopping rule was reached (154 female; $M_{age} = 23.02$, $SD = 3.81$, range = 18-44). Two participants were excluded from analyses because they reported one or more of the excluding criteria as specified in the preregistration (i.e., participants would be excluded if their Likert scale ratings on *Liking* and *Being Moved* how a monotonous pattern, if the time spent with each artwork image > 3 standard deviations (SD) than the average, and/or if participants completed the survey > 3 SDs from average survey completion time for the whole sample). We also excluded one participant who showed a monotonous answer pattern tested via standard deviations and Mahalanobis Distance larger than the critical chi-square value for dfs at $\alpha = .001$ (Mahalanobis, 1936), and one participant who spent longer than three SDs above mean total viewing time per artwork, leading to the final sample.

Stimuli

The same 60 artworks (30 representational, 30 abstract) from Study 1 were employed.

Procedure

The study was implemented and administered in *Qualtrics*. First, participants read the instructions, which summarized all parts of the study. The first task was to view and rate the artworks on four dimensions (*Liking*, *Being Moved*, *Valence*, *Arousal*). The artworks were not shown in a block design (abstract/representational) as in Study 1, but in a randomized order, while the ratings remained in the same order below each artwork. This was done to exclude the explanation that the block design (used in Study 1) might have influenced participants' ratings. The stimuli were presented at the center of the screen, with a maximum of 600 px width or height respectively of the landscape or portrait format and time was recorded on how long participants spent with each artwork. Thereafter, participants filled out the self-report questionnaires, which were displayed one at a time. No questionnaires or tasks other than those reported in this section were administered. The study lasted for about 45 min.

Artwork Ratings

All artworks were rated in four dimensions. We assessed *Liking* (“*How much do you like this artwork?*”) and *Being Moved* (“*How strongly does this artwork move you?*”). We also assessed *Valence* and *Arousal* which represent two main measures of affective states within the circumplex model of affect (Russel, 1980; Posner et al., 2005) and are commonly employed to assess emotional responses during art experience (e.g., Brieber et al., 2014; Dunn et al., 2010; Gerger et al., 2018). Arousal was assessed by asking “*How arousing is this artwork (i.e., how intense is the feeling/emotion that it triggers in you)?*” (1, *low*; 7, *high*) following Azevedo et al. (2020). Valence was assessed on a 7-point Likert scale by asking “*How positive or negative is this artwork?*” (1, *negative*; 7 *positive*). The ratings were displayed in the same order below each artwork: *Valence*, *Arousal*, *Being Moved*, and *Liking*.

Questionnaires

Interoceptive Sensitivity. We measured *IS* using The Multidimensional Assessment of Interoceptive Awareness, Version II (MAIA-2; Mehling et al., 2018). The MAIA-2 is a state-trait questionnaire to measure multiple dimensions of interoception by self-report. It consists of 37 statements which the participants rated on a 6-point Likert (0, *never*; 5, *always*). The total score, obtained by summing and averaging all items ranges from 0 to 5, with higher scores reflecting greater *IS*. The MAIA-2 has a high internal consistency reliability with Cronbach's α scores ranging from 0.64 to 0.83 across the eight scales (Mehling et al., 2018), described in the Supplementary Materials. In the current sample, Cronbach's α indices of the MAIA-2 subscales ranged from 0.67 to 0.85.

Need for Affect. We measured individual differences in attitudes toward feeling emotions and the tendency to approach or avoid emotion-inducing situations and activities using the short (NAQ-S, Appel et al., 2012) version of the Need for Affect Questionnaire (NAQ; Maio & Esses, 2001) which includes 10 out of the original 26 items. Participants responded to each of the statements using a 7-point scale (-3 , *strongly disagree*; 3 *strongly agree*). In the current sample, Cronbach's α was 0.61.

Aesthetic Experiences Scale. Afterward, the Aesthetic Experiences Scale (Silvia & Nusbaum, 2011) was administered to gain insight into the propensity to which our sample feels intense aesthetic experiences when engaging with art. First, participants were asked to write down the art domain they encounter most frequently in their life and then rated ten items, on a 7-point (1, *not at all*; 7 *very much*), on how much they experience feelings associated with feelings being touched, immersed, absorbed and having chills, among others, when they experience their most frequent art domain. Cronbach's α for the total score was 0.69 (*Chills* = 0.83; *Touched* = 0.67; *Absorbed* = 0.85).

Vicarious Pain. To measure reports and characteristics of conscious vicarious pain experiences, we employed the Vicarious Pain Questionnaire (VPQ; Grice-Jackson et al.,

2017), which consists of 16 video short (10-13 s) clips depicting injections ($N = 8$) and sports injuries ($N = 8$). After each video, participants report whether it triggered pain on their own body (giving a summed score across all videos from 0 to 16) by responding to the question “*Did you experience any bodily sensation of pain while observing the (e.g. arm injection)?*”. All participants were also asked to rate “*How unpleasant did you find the experience of watching this video?*” on a 10-point scale (1, *not unpleasant*; 10 *highly unpleasant*). Upon giving an affirmative answer they are then asked follow-up questions namely: to rate the intensity (on a scale ranging from 0 to 10), to select from a series of pain descriptors that describe sensory and affective qualities of pain, and to indicate whether the pain is local or general. This method allows us to separate participants into our two groups of interest: Non-responders (participants who report no pain when watching a video with someone else experiencing physical pain) and Vicarious Pain responders (VP-responders).²

Additional Demographics

As in Study 1, in addition to age and gender, we collected additional demographic information. Specifically, we asked questions concerning art education (yes or no - if yes, specify degree), art knowledge (with 6 questions concerning knowledge of iconography and production methods and a task to name the artists and styles of 10 artworks; scores can range from 0 to 26 for art knowledge) and art interest (11 questions rated on a 7-point scale, with the subjective interest questions having anchors from 1 (*not at all*) to 7 (*completely*) and the questions about concrete behaviors are rated on a scale from 1 (*less than once a year*) to 7 (*once a week or more often*); scores can range from 11 to 77) (assessed via VAIK; Specker et al., 2020), and art preference (representational or abstract). Finally, we asked participants whether they were familiar with the presented artworks (from *none* to *most of them*).

² We also employed the Mirror-Touch Questionnaire (MTQ; Li et al., 2022) but did not analyze it for the present study.

Results

Similarly to Study 1, more participants (68.91%) reported to prefer representational art than abstract art (30.18%)³. Regarding art interest and knowledge, participants showed a mean below the middle of the art interest scale ($M = 40.00$, $SD = 13.85$, $range = 12-74$), and just below the middle of the art knowledge scale ($M = 6.22$, $SD = 4.20$, $range = 0-25$) of the VAIK, with a positive correlation between both, $r(220) = .43$, $p < .001$, 95% CI [.32 .53]. As in Study 1, most participants (97.9%) had never undertaken any art training (see Supplementary Material Table S3 for full survey results).

Inter-Rater Agreement in Emotional Response and Appraisal

As with Study 1, we investigated inter-rater agreement across our artworks on the measured DVs. The inter-rater reliability of the ratings *Being Moved* ($kappa = 0.21$, $p < .001$), *Arousal* ($kappa = 0.22$, $p < .001$), and *Liking* ($kappa = 0.14$, $p < .001$) were minimal across participants, similarly to our Study 1 findings of highly individualized responses to our artwork set. The inter-rater reliability of the *Valence* ($kappa = 0.57$, $p < .001$) was weak across participants.

Effect of Art Type on Emotional Response and Liking

As per the preregistration, and matching Study 1, we conducted mixed-effect model analyses to address our research questions. To test the effect of artwork type on subjective ratings, we ran four different mixed models for each subjective rating (*Arousal*, *Being Moved*, *Liking*, *Valence*), employing the same model structure as in Study 1. The effect of art type was significant for *Arousal*, $F_{1, 57.99} = 24.67$, $p < .001$, and *Being Moved*, $F_{1, 58} = 41.25$, $p < .001$. Representational artworks were rated as higher in *Arousal* ($M = 4.48$, $SE = 0.14$) than abstract artworks ($M = 3.60$, $SE = 0.14$). Similarly, representational artworks were more

³ The analysis whether subjective ratings differ depending on art preference is reported in the Supplementary Material.

moving ($M = 4.08$, $SE = 0.22$) than abstract ones ($M = 3.02$, $SE = 0.13$), see Figure 7. We found no significant effect on *Valence* ratings, $F_{1, 58} = 1.36$, $p = .25$ and a marginally significant effect on *Liking*, $F_{1, 58} = 3.77$, $p = .057$.

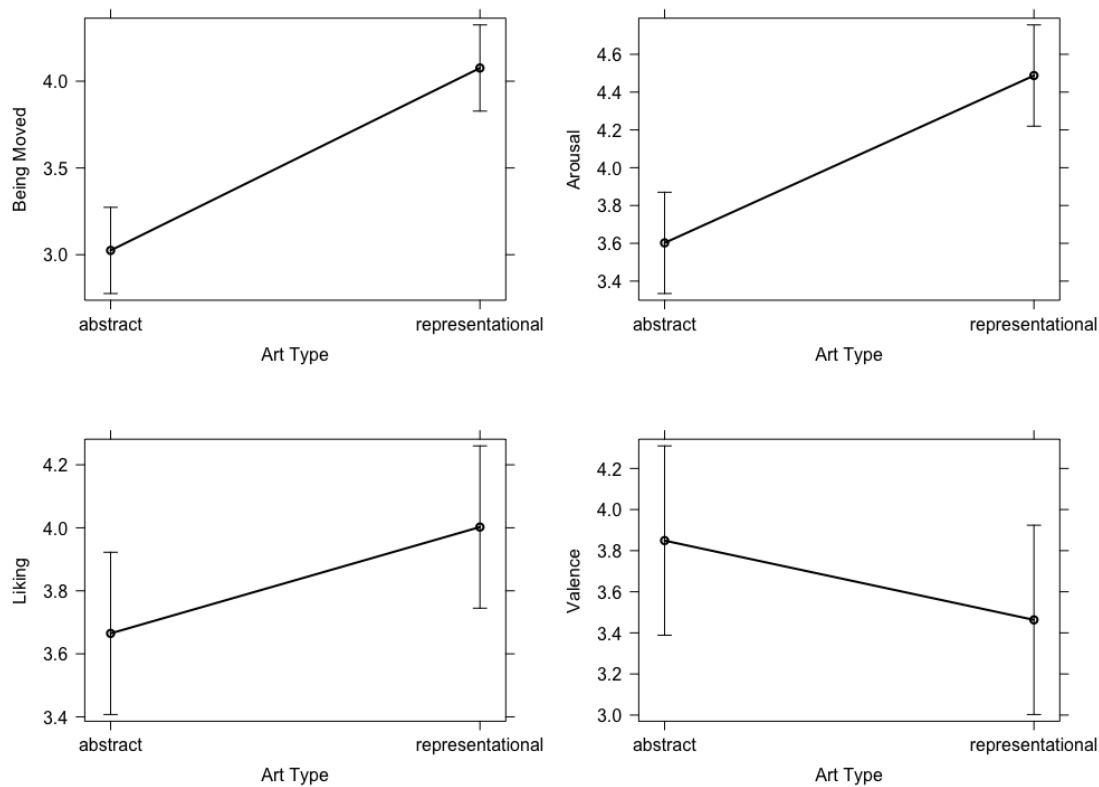


Figure 7. Effect of art type (abstract vs. representational) on the four artwork ratings. *Arousal* and *Being Moved* ratings were significantly ($p < .05$) higher for representational artworks.

Relationship Between Ratings of Art Experience

To test the relationship between different measures of appraisal and emotional response (*Arousal*, *Valence*, *Liking*, *Being Moved*) irrespective of art type, we used repeated measures correlations with R package *rmmcorr* (Bakdash & Marusich, 2017), see Table 1. In general, we found a positive relationship between *Being Moved* and *Liking* (which we note replicated in Study 1's results) as well as between these ratings and *Arousal*. *Valence* was negatively related to *Being Moved* and arousal, but positively related to *Liking*.

Table 1. Means, standard deviations, and correlations with confidence intervals for the four artwork ratings.

Variable	<i>M</i>	<i>SD</i>	1	2	3
1. liking	3.83	1.84			
2. valence	3.66	1.69	.32** [.30, .33]		
3. being moved	3.56	1.76	.47** [.45, .48]	-.13** [-.15, -.12]	
4. arousal	4.04	1.73	.34** [.33, .36]	-.20** [-.22, -.18]	.69** [.68, .70]

Note. *M* and *SD* are used to represent mean and standard deviation, respectively. Values in square brackets indicate the 95% confidence interval for each correlation. The confidence interval is a plausible range of population correlations that could have caused the sample correlation (Cumming, 2014). ** indicates $p < .01$.

Effect of Individual Differences in *IS* on Emotional Response and Appraisal

To assess whether individual differences in *IS* (assessed via the MAIA-2; Mehling et al., 2018) connected to the artwork reports, we used a grand average of all MAIA-2 subscales to assess overall *IS*. Our sample was characterized by a mean of 1.88 ($SD = 0.58$), which is relatively lower than other studies using this measure (e.g., Benau et al., 2023; Fanghella et al., 2022; Kühnapfel et al., 2023b; Millman et al., 2023; Ware et al., 2013), but still with a wide range (1.33 - 4.26).

First, we tested whether participants with higher *IS* exhibited higher levels of emotional response and appraisal. In contrast with our expectations, *IS* did not have a significant effect on any of the four ratings: *Liking*, $F_{1, 219} = 0.72$, $p = .40$, *Being Moved*, $F_{1, 219} = 0.14$, $p = .70$, *Arousal*, $F_{1, 219} = 0.20$, $p = .65$, or *Valence*, $F_{1, 219} = 1.50$, $p = .22$.

Second, we expected that the effects of *IS* on ratings would be stronger for representational compared to abstract artworks. We found a significant interaction between art type and *IS* only on *Valence*, $F_{1, 12979} = 10.09$, $p = .001$. The relationship between *IS* and *Valence* was stronger and more positive for representational art ($trend = 0.01$, $SE = 0.04$) compared to abstract art ($trend = -0.01$, $SE = 0.04$). No significant interaction was found for other variables (*Liking*: $F_{1, 12979} = 0.24$, $p = .62$; *Being Moved*: $F_{1, 12979} = 0.42$, $p = .42$; *Arousal*, $F_{1, 12979} = 2.18$, $p = .14$).

Exploratory Analysis: MAIA-2 Subscales Emotional Response and Appraisal

As an exploratory follow-up (i.e., expanding beyond our initial pre-registration), the individual effect of MAIA-2 subscales on subjective ratings was subsequently assessed (see Supplementary Materials for descriptive statistics on the subscales). We note that due to the exploratory nature of these analyses, all significance levels for correlations are reported uncorrected with alpha set at 0.05 (see Armstrong, 2014; Cox & Van Klaveren, 2021).

For this analysis, we specifically focused on the five subscales that assess the ability to perceive and “listen to” physiological changes inside one’s body, which we considered to capture awareness of the body and physiological sensibility most adequately (i.e., *Noticing*, *Emotional Awareness*, *Body Listening*, *Attention Regulation*, and *Trusting* scales) (Ventura-Bort et al., 2021).

Emotional Awareness scores were positively associated with *Arousal* ratings, $b = 0.14$, $F_{1, 208} = 4.83$, $p = .03$, and *Being Moved* ratings, $b = 0.18$, $F_{1, 208} = 7.66$, $p = .006$. Similarly, individuals who scored high on *Body Listening* subscale reported higher *Arousal*, $b = 0.11$, $F_{1, 220.1} = 5.58$, $p = .029$. Individual differences in *Noticing*, *Attention Regulation*, and *Trusting* did not affect artwork ratings. The type of artwork did not influence the relationship between *Body Listening* and the ratings (*Being Moved*: $F_{1, 13038} = 2.88$, $p = .90$; *Arousal*: $F_{1, 13038} = 1.03$, $p = .31$), but it influenced the relationship between *Emotional Awareness* and

Being Moved, $F_{1, 12330} = 7.37, p = .007$, as well as *Arousal*, $F_{1, 12330} = 6.49, p = .01$.

Specifically, the relationship between *Emotional Awareness* and these ratings were stronger for representational (*Being Moved*: $trend = 0.22, SE = 0.07$; *Arousal*: $trend = 0.18, SE = 0.07$) than abstract artworks (*Being Moved*: $trend = 0.14, SE = 0.07$; *Arousal*: $trend = 0.11, SE = 0.07$).

Effect of Individual Differences in Emotion Scales on Art Experience

Need for Affect

Need for Affect had a significant effect on *Valence*, $b = 0.06, F_{1, 220} = 4.43, p = .04$.

There was no significant effect on *Being Moved*, $b = 0.04, F_{1, 220} = 0.23, p = .62$, *Arousal*, $b = 0.06, F_{1, 220} = 0.72, p = .59$, and *Liking*, $b = 0.08, F_{1, 220} = 1.51, p = .22$.

Aesthetic Experiences Scale

The total AES scale predicted a significant increase in *Being Moved*, $b = 0.36, F_{1, 220} = 74.61, p < .001$, $b = 0.37, F_{1, 220} = 63.80, p < .001$, and *Liking*, $b = 0.22, F_{1, 220} = 28.23, p < .001$, but did not predict *Valence*, $F_{1, 220} = 0.01, p = .93$.

Vicarious Pain

Participants were categorized into one of three responder groups: Non-responders (controls), Sensory-Localised responders, and Affective-Generalised responders, following the same two-step cluster analysis described in Grice-Jackson et al. (2017) (see Supplementary Materials for the full procedure). For the scope of this paper, we considered the difference in art experience between “VP-responders” (who report experiencing conscious vicarious experiences) and “Non-responders” (control group). The final sample ($N = 222$) comprised $N = 89$ Non-responders (61 female; $M_{age} = 22.97, SD = 3.12$, age range 18-32) and $N = 133$ VP-responders (93 female; $M_{age} = 23.02, SD = 4.24$, age range 18-44).

We next examined whether art ratings would differ between groups. We found that *Being Moved*, $F_{1, 220} = 11.59, p < .001$, and *Arousal*, $F_{1, 220} = 13.04, p < .001$, significantly

differed between the control and VP-responders. Post hoc comparisons showed that VP-responders had higher means for *Being Moved* ($M = 3.69$, $SE = 0.12$) than Non-responders ($M = 3.34$, $SE = 0.13$). VP-responders also provided higher *Arousal* ratings ($M = 4.20$, $SE = 0.13$) than Non-responders ($M = 3.81$, $SE = 0.13$). There was no significant difference for *Valence*, $F_{1,220} = 0.37$, $p = .54$, and *Liking*, $F_{1,220} = 0.22$, $p = .64$.

Discussion Study 2

While Study 1 focused on an objective indicator of interoception (*IAC*), Study 2 investigated a subjective measure pertaining to participants' beliefs regarding their interoceptive abilities (*IS*). Here, higher scores on *IS* were found to predict more positive valence for representational art, while no effect of this trait variation was observed for other subjective ratings. However, exploratory analyses on the subscales of the *MAIA* questionnaire revealed that individual differences in the emotion-body connection (*Emotional Awareness*) and in gaining insight from the body by actively listening to the body (*Body Listening*) predicted emotional response to art (particularly those pertaining artworks' emotional intensity, e.g., *Being Moved* or *Arousal*). Moreover, individuals high in *Emotional Awareness* rated representational artworks to be more moving and arousing.

Concerning individual differences in emotion, we observed a complex pattern of results. Specifically, Need for Affect was positively related to valence ratings. Future research would be needed to establish a relationship between positive bias in affective processing and the Need for Affect scale. In contrast, individual differences in the tendency for emotional/bodily art experiences (measured via AES) predicted *Arousal*, *Liking*, and *Being Moved*, but not *Valence* ratings. So far, the AES has only been employed with regard to a link to individual differences in openness to experience: this trait has been associated with the prevalence of the kind of experiences in the AES (Silvia & Nusbaum, 2011; Silvia et al., 2015) or in an adapted version to rate artworks (Specker et al., 2017) such that we provide

the first link between AES and art experience. Finally, we found, for the first time, that people who report having vicarious pain experiences (VP responders), reported higher ratings of *Arousal* and *Being Moved* towards both kinds of art compared to non-responders.

Study 2 also replicated the finding from Study 1 that emotional experience (*Being Moved*, *Arousal*) is more intense for representational compared to abstract art. Importantly, we did not find that the two types of artworks differed in valence, suggesting that our stimulus set is balanced on this dimension. Hence, albeit this data was collected on a different (but larger) sample, we can cautiously exclude the possibility that *HR* deceleration observed in Study 1 for representational artworks was due to the differences in valence (see Figure A1 and A2 in Appendix).

General Discussion

Across two studies, we combined physiological, behavioral and questionnaire measures to investigate how emotional processing and individual differences in interoception contribute to emotional response and appraisals of visual art. Overall, our results establish the relevance of embodied accounts of art perception and evaluation, while also demonstrating an interplay between observer traits, emotional responses and artwork properties.

In both studies, for the first time, we found a link between individual differences in interoception and art experience ratings of emotional strength/intensity (*Arousal*, *Being Moved*), however, this relationship was mainly observed for representational artworks. Specifically, in Study 1, *IAc* was associated with higher *Being Moved* ratings, whereas in Study 2, *IS* was associated with more positive *Valence*. Additionally, *IS* subscales of *Emotional Arousal* and *Body Listening* were related to ratings of emotional intensity of representational artworks. The fact that we observed a role of interoceptive and emotional individual differences only for representational art is not entirely surprising, since as we discussed in the previous sections, this type of art conveys emotions in a more direct manner

than abstract art, for which art appraisal may be affected by other factors (e.g., viewer expertise, van Paasschen et al., 2015, or other aspects of personality or thinking style, Gridley, 2013).

We also found that other dimensions linked to interoception predicted art ratings. Specifically, in Study 2 we found for the first time that VP-responders, a population that is typically high in interoceptive abilities (Bowling et al., 2019), empathy (e.g., Lockwood et al., 2016; Young et al., 2017), and heightened emotional reactivity (Botan et al., 2017) provided higher ratings of *Arousal* and *Being Moved*, independently of art type, supporting a more general role of this individual trait in the emotional appraisal of art. Overall, our results suggest that bodily/emotional/interoceptive individual differences are involved with emotional appraisal of art, particularly the intensity of emotion, that may tap into similar dimensions, rather than with cognitive appraisal ratings (e.g., *Liking*). We note that AES was positively related to *Liking* in addition to *Arousal* and *Being Moved* ratings, which may be due to this trait being related to the propensity to feel intense emotions and bodily experiences (e.g., chills) when specifically engaging with art.

In terms of the role of physiological responses in art experience, we found a complex pattern of results. *HR* deceleration was more pronounced for representational art, and modulated *Being Moved* ratings only for this type of art. However, we found that *SCR* directly predicted both appraisal and emotion ratings. This is the first time that a direct link between bodily responses and art experience was found. This finding supports theoretical arguments of embodied art appraisal theories (Fingerhut & Prinz, 2018; also see Kühnapfel et al., 2023a) building on embodied appraisal theories of emotions (Prinz, 2004, also see accounts of embodied affectivity, Colombetti, 2014; Fuchs & Koch, 2014). It remains an interesting question whether the observed bodily signals are constitutive of art experience

(playing a causal role) or rather just accompanying experiences (see Casile & Ticini, 2014 for a discussion).

We also found a positive relationship between ratings of art appraisal (*Liking*) and emotional response across both studies (*Being Moved* in Study 1 and 2 and *Arousal* in Study 2). This, together with the impact of physiological/bodily responses (*HR*, *SCR*) on the ratings, and the interaction that emerged between *IAC* and *SCR* in predicting *Liking* observed in Study 1, provides evidence in support of the view that we use emotions as information when appraising art (i.e., aesthetic sentimentalism, Prinz, 2007; 2011; 2014; Prinz & Fingerhut, 2018; 2020). Accordingly, any aesthetic evaluation could be seen as an emotional process, in that we base our aesthetic judgment on the emotional pull and embodied cognitive style that certain emotions afford (for reviews see Fingerhut & Kühnapfel, *forthc.* Fingerhut & Prinz, 2020; Fingerhut & Spee, *forthc.*).

Last, in both studies, we observed that ratings underlying participants' art experience, albeit highly individualized, differed between representational and abstract artworks, with the former eliciting stronger emotions (not valence) and higher appraisal. This is in line with Gerger et al. (2018), who found a similar result when comparing emotional ratings (arousal and valence) towards abstract and representational artworks. We also note that our samples of participants were mostly naïve to art. There is evidence that representational art is preferred over abstract art among laypeople (this is the case for our sample too), while abstract art is often preferred by art experts and artists (Van Paasschen et al., 2015). However, both categories of participants (expert and non-expert) tend to show coherent emotional reactions to artworks (Van Paasschen et al., 2015). The higher appreciation found for representational paintings could also be due to their more could also be due to the different properties of the two types of artworks: representational paintings are more explicit regarding their semantic content, and shared semantic interpretations (Schepman et al., 2015; 2021) can lead to shared

preferences and thus provoke more agreement in appraisal ratings (Vessel & Rubin, 2010; Schepman et al., 2015).

Our studies are not without limitations. Heartbeat tracking paradigms, as used in Study 1, have been criticized for being confounded with expectations regarding one's heart rate (e.g., Ring et al., 2015), differ depending on the individual (i.e., health, fitness) or daytime, and people generally underestimating their own heart rate (Legrand et al., 2022; for a recent review, see Murphy 2023). We note that although HPT has been so far the most used measure to index *IAc* (Critchley & Garfinkel, 2017), future studies should look into other measures of interoceptive accuracy that have been recently developed. For instance, new interoception tasks include the Heartbeat Matching task, whereby participants use a slider to adjust the rate at which a heart icon is pulsing on a PC screen (Palmer et al., 2023), or the Phase Adjustment Taks (PAT; Plans et al., 2021) in which participants adjust the phase between tones and heartbeats until they are synchronous (for review on recent developments in interoception measures see Desmedt et al., 2022; 2023). Additionally, assessing *IAc* by combining the heartbeat-based tasks with other modalities such as respiratory resistance and breathing-related interoception may represent an interesting future direction for assessing a more comprehensive account of individual variation in interoception (Daubenmier et al., 2013; Garfinkel et al., 2016; Harrison et al., 2021).

On the other hand, subjective interoceptive sensitivity measures individual differences in perceived sensitivity across a range of internal bodily changes but does not address whether this sensibility is accurate or corresponds to objective measures of interoceptive accuracy (Garfinkel et al., 2016). However, survey-based self-report measures are easier to administer and require less time for completion. In addition, these measures allow studies to move from the lab to more ecological settings, which simulates the context where art is usually experienced (i.e., not as reproductions on a screen, but in its original form in art

museums/galleries) as it has been heavily demanded by empirical aesthetics scholars (e.g., see Brieber et al., 2015; Pelowski et al., 2016; Carbon, 2020). Besides behavioral tests of accuracy and self-report measures, more recent literature suggests incorporating interoceptive attention and higher-order measures related to the interpretation of interoceptive signals to capture the complexity of interoception (e.g., Suksasilp & Garfinkel, 2022). In alternative to assessing individual differences in interoceptive abilities, to investigate the causal role of interoception, future studies could also employ interoception training or interventions (e.g., Quadt et al., 2021) or body awareness meditations (e.g., Dekezyer, 2022) before the art engagement to test the impact of trained interoception on art experience.

To conclude, we provide novel evidence on the role of individual differences in interoception and emotion in the experience of visual art. This link mainly emerged with respect to the intensity of emotional experience in art, and more consistently with representational art, which may convey emotions more directly and impactfully than abstract art. In addition, ratings reflecting art experience were generally higher for representational art, potentially indicating more proactive introspecting emotions with such works. We also show that stronger bodily responses to art (as indexed by *SCR*) were linked to higher appraisal and being moved ratings. As such, our findings provide general support for aesthetic sentimentalism and embodied accounts of art and aesthetic experience. More research will be needed to elucidate this link with further measures of interoceptive abilities and their relation to art experiences and evaluations.

References

- Appel, M., Gnambs, T., & Maio, G. R. (2012). A short measure of the Need for Affect. *Journal of Personality Assessment*, 94(4), 418–426.
<https://doi.org/10.1080/00223891.2012.666921>
- Armstrong, R. A. (2014). When to use the Bonferroni correction. *Ophthalmic and Physiological Optics*, 34(5), 502–508. <https://doi.org/10.1111/opo.1213>
- Arslanova, I., Galvez-Pol, A., Kilner, J., Finotti, G., & Tsakiris, M. (2022). Seeing through each other's hearts: Inferring others' heart rate as a function of own heart rate perception and perceived social Intelligence. *Affective Science*, 3(4), 862–877.
<https://doi.org/10.1007/s42761-022-00151-4>
- Azevedo, R., Tucciarelli, R., De Beukelaer, S., Ambroziak, K., Jones, I., & Tsakiris, M. (2020). *A body of evidence: 'Feeling in seeing' predicts realness judgments for photojournalistic images*. [Preprint]. PsyArXiv. <https://doi.org/10.31234/osf.io/p7kan>
- Azzalini, D., Rebollo, I., & Tallon-Baudry, C. (2019). Visceral Signals Shape Brain Dynamics and Cognition. *Trends in Cognitive Sciences*, 23(6), 488–509.
<https://doi.org/10.1016/j.tics.2019.03.007>
- Bagby, R. M., Parker, J. D. A., & Taylor, G. J. (1994). The twenty-item Toronto Alexithymia scale—I. Item selection and cross-validation of the factor structure. *Journal of Psychosomatic Research*, 38(1), 23–32. [https://doi.org/10.1016/0022-3999\(94\)90005-1](https://doi.org/10.1016/0022-3999(94)90005-1)
- Bakdash, J.Z. & Marusich, L.R. (2017). Repeated Measures Correlation, *Frontiers in Psychology*, 8, 456, <https://doi.org/10.3389/fpsyg.2017.00456>
- Barrett, L. F., Quigley, K. S., Bliss-Moreau, E., & Aronson, K. R. (2004). Interoceptive sensitivity and self-reports of emotional experience. *Journal of Personality and Social Psychology*, 87(5), 684–697. <https://doi.org/10.1037/0022-3514.87.5.684>

- Bartsch, A., Appel, M., & Storch, D. (2010). Predicting emotions and meta-emotions at the movies: The role of the need for affect in audiences' experience of horror and drama. *Communication Research*, 37(2), 167–190. <https://doi.org/10.1177/0093650209356441>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2014). Fitting linear mixed-effects models using lme4. arXiv preprint arXiv:1406.5823. <https://doi.org/10.48550/arXiv.1406.5823>
- Benau, E. M. (2023). Self-reported interoceptive accuracy and interoceptive attention differentially correspond to measures of visual attention and self-regard. *PeerJ*, 11, e15348. <https://doi.org/10.7717/peerj.1534>
- Berlyne, D. E. (1971). *Aesthetics and psychobiology*. New York: Appleton-Century-Crofts.
- Botan, V., Bowling, N. C., Banissy, M. J., Critchley, H., & Ward, J. (2018). Individual differences in vicarious pain perception linked to heightened socially elicited emotional states. *Frontiers in Psychology*, 9, 2355. <https://doi.org/10.3389/fpsyg.2018.02355>
- Bowling, N. C., Botan, V., Santiesteban, I., Ward, J., & Banissy, M. J. (2019). Atypical bodily self-awareness in vicarious pain responders. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 374(1787), 20180361. <https://doi.org/10.1098/rstb.2018.0361>
- Bradley, M. M., & Lang, P. J. (2007). The International Affective Picture System (IAPS) in the study of emotion and attention. In J. A. Coan & J. J. B. Allen (Eds.), *Handbook of emotion elicitation and assessment* (pp. 29–46). London: Oxford University Press.
- Brewer, R., Cook, R., & Bird, G. (2016). Alexithymia: A general deficit of interoception. *Royal Society Open Science*, 3(10), 150664. <https://doi.org/10.1098/rsos.150664>
- Brieber, D., Nadal, M., & Leder, H. (2015). In the white cube: Museum context enhances the valuation and memory of art. *Acta Psychologica*, 154, 36–42. <https://doi.org/10.1016/j.actpsy.2014.11.004>

- Briellmann, A. A., & Pelli, D. G. (2019). Intense beauty requires intense pleasure. *Frontiers in Psychology*, 10, 2420. <https://doi.org/10.3389/fpsyg.2019.02420>
- Brown, S., Gao, X., Tisdelle, L., Eickhoff, S. B., & Liotti, M. (2011). Naturalizing aesthetics: Brain areas for aesthetic appraisal across sensory modalities. *NeuroImage*, 58(1), 250–258. <https://doi.org/10.1016/j.neuroimage.2011.06.012>
- Carbon, C.-C. (2020). Ecological art experience: How we can gain experimental control while preserving ecologically valid settings and contexts. *Frontiers in Psychology*, 11, 800. <https://doi.org/10.3389/fpsyg.2020.00800>
- Casile, T., & Ticini, L.F. (2014). The role of sensory and motor systems in art appreciation and implications for exhibit design. In N. S. Levent, & A. Pascual-Leone (Eds.), *The Multisensory Museum: Cross-Disciplinary Perspectives on Touch, Sound, Smell, Memory, and Space* Rowman & Littlefield.
- Chapman, L. J., Chapman, J. P., & Raulin, M. L. (1976). Scales for physical and social anhedonia. *Journal of Abnormal Psychology*, 85(4), 374–382. <https://doi.org/10.1037/0021-843X.85.4.374>
- Chatterjee, A., & Vartanian, O. (2014). Neuroaesthetics. *Trends in Cognitive Sciences*, 18(7), 370–375. <https://doi.org/10.1016/j.tics.2014.03.003>
- Christensen, J. F., Azevedo, R. T., & Tsakiris, M. (2021). Emotion matters: Different psychophysiological responses to expressive and non-expressive full-body movements. *Acta Psychologica*, 212, 103215. <https://doi.org/10.1016/j.actpsy.2020.103215>
- Colombetti, G. (2014). *The feeling body: Affective science meets the enactive mind*. Cambridge, Massachusetts: MIT press.
- Cox, R. F. A., & van Klaveren, L.-M. (2022). The embodied experience of abstract art: An exploratory study [Preprint]. Open Science Framework. <https://doi.org/10.31219/osf.io/s4nry>

- Craig, A. D. (2002). How do you feel? Interoception: The sense of the physiological condition of the body. *Nature Reviews Neuroscience*, 3(8), Article 8.
<https://doi.org/10.1038/nrn894>
- Critchley H. D. (2002). Electrodermal responses: What happens in the brain. *The Neuroscientist: A review journal bringing neurobiology, neurology and psychiatry*, 8(2), 132–142. <https://doi.org/10.1177/107385840200800209>
- Critchley, H. D., & Harrison, N. A. (2013). Visceral influences on brain and behavior. *Neuron*, 77(4), 624–638. <https://doi.org/10.1016/j.neuron.2013.02.008>
- Critchley, H. D., Wiens, S., Rotshtein, P., Ohman, A., & Dolan, R. J. (2004). Neural systems supporting interoceptive awareness. *Nature Neuroscience*, 7(2), 189–195.
<https://doi.org/10.1038/nrn1176>
- Cupchik, G. C. (2016). *The aesthetics of emotion: Up the down staircase of the mind-body* (1st ed.). Cambridge, UK: Cambridge University Press.
<https://doi.org/10.1017/CBO9781139169301>
- Cupchik, G. C., Vartanian, O., Crawley, A., & Mikulis, D. J. (2009). Viewing artworks: Contributions of cognitive control and perceptual facilitation to aesthetic experience. *Brain and Cognition*, 70(1), 84–91. <https://doi.org/10.1016/j.bandc.2009.01.003>
- Damasio, A., & Carvalho, G. B. (2013). The nature of feelings: Evolutionary and neurobiological origins. *Nature Reviews Neuroscience*, 14(2), Article 2.
<https://doi.org/10.1038/nrn3403>
- Daubenmier, J., Sze, J., Kerr, C. E., Kemeny, M. E., & Mehling, W. (2013). Follow your breath: Respiratory interoceptive accuracy in experienced meditators: Respiratory interoceptive accuracy in meditators. *Psychophysiology*, 50(8), 777–789.
<https://doi.org/10.1111/psyp.12057>

- Dekeyzer, B. (2022). From within the body: Proprioceptive art experiences in the museum. *Journal of Visual Literacy*, 41(1), 29–45.
<https://doi.org/10.1080/1051144X.2021.1994731>
- Desmedt, O., Heeren, A., Corneille, O., & Luminet, O. (2022). What do measures of self-report interoception measure? Insights from a systematic review, latent factor analysis, and network approach. *Biological Psychology*, 169, 108289.
<https://doi.org/10.1016/j.biopsycho.2022.108289>
- Desmedt, O., Luminet, O., Walentynowicz, M., & Corneille, O. (2023). *The new measures of interoceptive accuracy: A systematic review and assessment* [Preprint]. Open Science Framework. <https://doi.org/10.31219/osf.io/f56xe>
- Donges, U.-S., Kersting, A., & Suslow, T. (2014). Alexithymia and perception of emotional information: A review of experimental psychological findings. *Universitas Psychologica*, 13(2), 745–756. <https://doi.org/10.11144/Javeriana.UPSY13-2.apei>
- Dunn, B. D., Galton, H. C., Morgan, R., Evans, D., Oliver, C., Meyer, M., Cusack, R., Lawrence, A. D., & Dalgleish, T. (2010). Listening to your heart. How interoception shapes emotion experience and intuitive decision making. *Psychological Science*, 21(12), 1835–1844. <https://doi.org/10.1177/0956797610389191>
- Duschek, S., Werner, N. S., Reyes Del Paso, G. A., & Schandry, R. (2015). The contributions of interoceptive awareness to cognitive and affective facets of body experience. *Journal of Individual Differences*, 36(2), 110–118. <https://doi.org/10.1027/1614-0001/a000165>
- Eggart, M., Lange, A., Binser, M., Queri, S., & Müller-Oerlinghausen, B. (2019). Major depressive disorder is associated with impaired interoceptive accuracy: A systematic review. *Brain Sciences*, 9(6), 131. <https://doi.org/10.3390/brainsci9060131>

Ehlers, A. (1998). Why Van der Does, Van Dyck and Spinhoven (1997) did not demonstrate that superior heartbeat perception in panic disorder is partly an artefact. *Journal of Affective Disorders*, 49(3), 241–247.

Fanghella, M., Gaigg, S. B., Candidi, M., Forster, B., & Calvo-Merino, B. (2022).

Somatosensory evoked potentials reveal reduced embodiment of emotions in autism. *The Journal of Neuroscience*, 42(11), 2298–2312.

<https://doi.org/10.1523/JNEUROSCI.0706-21.2022>

Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/BF03193146>

Fingerhut, J., & Kühnapfel, C. (forthc.). Art and embodied aesthetic emotions. A. Scarinzi (ed.). In “*Being Moved to Interest*”: *Interestingness, Emotions and 4E Cognition*

Fingerhut, J., & Prinz, J. J. (2018). Wonder, appreciation, and the value of art. In J. F. Christensen & A. Gomila (Eds.), *Progress in Brain Research* (Vol. 237, pp. 107–128). Elsevier. <https://doi.org/10.1016/bs.pbr.2018.03.004>

Fingerhut, J., & Prinz, J. J. (2020). Aesthetic emotions reconsidered. *The Monist*, 103(2), 223–239. <https://doi.org/10.1093/monist/onz037>

Fingerhut, J., & Spee, B. T. M. (forthc.). Aesthetic emotions, play, and radical predictive processing of art. *Philosophical Transactions of the Royal Society B: Biological Sciences*.

Fowles, D. C, Christie, M. J, Edelberg, R., Grings, W. W, Lykken, D. T, & Venables, P. H. (1981). Publication recommendations for electrodermal measurements. *Psychophysiology*, 18(3), 232–239. <https://doi.org/10.1111/j.1469-8986.1981.tb03024.x>

- Fuchs, T., & Koch, S. C. (2014). Embodied affectivity: On moving and being moved. *Frontiers in Psychology*, 5. <https://doi.org/10.3389/fpsyg.2014.00508>
- Füstös, J., Gramann, K., Herbert, B. M., & Pollatos, O. (2013). On the embodiment of emotion regulation: Interoceptive awareness facilitates reappraisal. *Social Cognitive and Affective Neuroscience*, 8(8), 911–917. <https://doi.org/10.1093/scan/nss089>
- Garfinkel, S. N., Manassei, M. F., Hamilton-Fletcher, G., In Den Bosch, Y., Critchley, H. D., & Engels, M. (2016). Interoceptive dimensions across cardiac and respiratory axes. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 371(1708), 20160014. <https://doi.org/10.1098/rstb.2016.0014>
- Garfinkel, S. N., Seth, A. K., Barrett, A. B., Suzuki, K., & Critchley, H. D. (2015). Knowing your own heart: Distinguishing interoceptive accuracy from interoceptive awareness. *Biological Psychology*, 104, 65–74. <https://doi.org/10.1016/j.biopsycho.2014.11.004>
- Gerger, G., Pelowski, M., & Leder, H. (2018). Erratum to: Empathy, Einfühlung, and aesthetic experience: The effect of emotion contagion on appreciation of representational and abstract art using fEMG and SCR. *Cognitive processing*, 19(2), 167. <https://doi.org/10.1007/s10339-017-0807-8>
- Giannini, A. M., Tizzani, E., Baralla, F., & Gurrieri, G. (2013). What I like is how I am: Impact of alexithymia on aesthetic preference. *Creativity Research Journal*, 25(3), 312–316. <https://doi.org/10.1080/10400419.2013.813796>
- Grice-Jackson, T., Critchley, H. D., Banissy, M. J., & Ward, J. (2017). Common and distinct neural mechanisms associated with the conscious experience of vicarious pain. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, 94, 152–163. <https://doi.org/10.1016/j.cortex.2017.06.015>
- Gridley, M. (2013). Preference for abstract art according to thinking styles & personality. *North American Journal of Psychology*, 15, 463–481.

- Harrison, O. K., Garfinkel, S. N., Marlow, L., Finnegan, S. L., Marino, S., Köchli, L., Allen, M., Finnemann, J., Keur-Huizinga, L., Harrison, S. J., Stephan, K. E., Pattinson, K. T. S., & Fleming, S. M. (2021). The Filter Detection Task for measurement of breathing-related interoception and metacognition. *Biological Psychology*, 165, 108185.
<https://doi.org/10.1016/j.biopsycho.2021.108185>
- Høffding, S., Sánchez, C. V., & Roald, T. (2022). Being moved by art: A phenomenological and pragmatist dialogue. *Estetika: The European Journal of Aesthetics*, LIX/XV(2), 85–102. <https://doi.org/10.33134/eeja.246>
- James, W. (1884). What is an Emotion? *Mind*, os-IX(34), 188–205.
<https://doi.org/10.1093/mind/os-IX.34.188>
- Jola, C., Pollick, F. E., & Calvo-Merino, B. (2014). “Some like it hot”: Spectators who score high on the personality trait openness enjoy the excitement of hearing dancers breathing without music. *Frontiers in Human Neuroscience*, 8.
<https://www.frontiersin.org/articles/10.3389/fnhum.2014.00718>
- Judd, C. M., Westfall, J., & Kenny, D. A. (2012). Treating stimuli as a random factor in social psychology: A new and comprehensive solution to a pervasive but largely ignored problem. *Journal of Personality and Social Psychology*, 103(1), 54–69.
<https://doi.org/10.1037/a0028347>
- Jung, W.-M., Ryu, Y., Lee, Y.-S., Wallraven, C., & Chae, Y. (2017). Role of interoceptive accuracy in topographical changes in emotion-induced bodily sensations. *PLOS ONE*, 12(9), e0183211. <https://doi.org/10.1371/journal.pone.0183211>
- Juslin P. N. (2013). From everyday emotions to aesthetic emotions: towards a unified theory of musical emotions. *Physics of life reviews*, 10(3), 235–266.
<https://doi.org/10.1016/j.plrev.2013.05.008>

- Kaltwasser, L., Hildebrandt, A., Wilhelm, O., & Sommer, W. (2017). On the relationship of emotional abilities and prosocial behavior. *Evolution and Human Behavior*, 38(3), 298-308. <https://doi.org/10.1016/j.evolhumbehav.2016.10.011>
- Kaltwasser, L., Moore, K., Weinreich, A., & Sommer, W. (2017). The influence of emotion type, social value orientation and processing focus on approach-avoidance tendencies to negative dynamic facial expressions. *Motivation and Emotion*, 41(4), 532-544. <https://doi.org/10.1007/s11031-017-9624-8>
- Kaltwasser, L., Rost, N., Ardizzi, M., Calbi, M., Settembrino, L., Fingerhut, J., Pauen, M., & Gallese, V. (2019). Sharing the filmic experience - The physiology of socio-emotional processes in the cinema. *PloS One*, 14(10), e0223259. <https://doi:10.1371/journal.pone.0223259>
- Kever, A., Pollatos, O., Vermeulen, N., & Grynberg, D. (2015). Interoceptive sensitivity facilitates both antecedent- and response-focused emotion regulation strategies. *Personality and Individual Differences*, 87, 20–23. <https://doi.org/10.1016/j.paid.2015.07.014>
- Kreibig S. D. (2010). Autonomic nervous system activity in emotion: a review. *Biological psychology*, 84(3), 394–421. <https://doi.org/10.1016/j.biopsycho.2010.03.010>
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest Package: Tests in Linear Mixed Effects Models. *Journal of Statistical Software*, 82, 1–26. <https://doi.org/10.18637/jss.v082.i13>
- Kühnapfel, C., Fingerhut, J., Brinkmann, H., Ganster, V., Tanaka, T., Specker, E., Mikuni, J., Gildenpfennig, F., Gartus, A., Rosenberg, R., & Pelowski, M. (2023a). How do we move in front of art? How does this relate to art experience? Linking movement, eye tracking, emotion, and evaluations in a gallery-like setting. *Empirical Studies of the Arts*, 027623742311600. <https://doi.org/10.1177/02762374231160000>

- Kühnapfel, C., Fingerhut, J., & Pelowski, M. (2023b). The role of the body in the experience of installation art: A case study of visitors' bodily, emotional, and transformative experiences in Tomás Saraceno's "in orbit". *Frontiers in Psychology*, 14, 1192689. <https://doi.org/10.3389/fpsyg.2023.1192689>
- Lang, P. J., Greenwald, M. K., Bradley, M. M., & Hamm, A. O. (1993). Looking at pictures: affective, facial, visceral, and behavioral reactions. *Psychophysiology*, 30(3), 261–273. <https://doi.org/10.1111/j.1469-8986.1993.tb03352.x>
- Leder, H., Belke, B., Oeberst, A., & Augustin, D. (2004). A model of aesthetic appreciation and aesthetic judgments. *British Journal of Psychology*, 95(4), 489–508. <https://doi.org/10.1348/0007126042369811>
- Leder, H., & Nadal, M. (2014). Ten years of a model of aesthetic appreciation and aesthetic judgments: The aesthetic episode - Developments and challenges in empirical aesthetics. *British Journal of Psychology*, 105(4), 443–464. <https://doi.org/10.1111/bjop.12084>
- Legrand, N., Nikolova, N., Correa, C., Brændholt, M., Stuckert, A., Kildahl, N., Vejlø, M., Fardo, F., & Allen, M. (2022). The heart rate discrimination task: A psychophysical method to estimate the accuracy and precision of interoceptive beliefs. *Biological Psychology*, 168, 108239. <https://doi.org/10.1016/j.biopsycho.2021.108239>
- Lenth, R. V., Bolker, B., Buerkner, P., Giné-Vázquez, I., Herve, M., Jung, M., Love, J., Miguez, F., Riebl, H., & Singmann, H. (2023). *emmeans: Estimated Marginal Means, aka Least-Squares Means* (1.8.5). <https://cran.r-project.org/web/packages/emmeans/index.html>
- Li, M., Hao, L., Ren, Z., Qiu, J., & Ward, J. (2022). Vicarious experiences of touch (mirror touch) in a Chinese sample: Cross-cultural and individual differences. *PloS one*, 17(11), e0266246. <https://doi.org/10.1371/journal.pone.0266246>

- Lockwood, P. L. (2016). The anatomy of empathy: Vicarious experience and disorders of social cognition. *Behavioural Brain Research*, 311, 255–266.
<https://doi.org/10.1016/j.bbr.2016.05.048>
- Maio, G. R., & Esses, V. M. (2001). The need for affect: Individual differences in the motivation to approach or avoid emotions. *Journal of Personality*, 69(4), 583–615.
<https://doi.org/10.1111/1467-6494.694156>
- Mahalanobis, P.C. (1936) On the Generalized Distance in Statistics. *Proceedings of the National Institute of Science of India*, 2, 49-55.
- Mastandrea, S., Maricchiolo, F., Carrus, G., Giovannelli, I., Giuliani, V., & Berardi, D. (2019). Visits to figurative art museums may lower blood pressure and stress. *Arts & Health*, 11(2), 123–132. <https://doi.org/10.1080/17533015.2018.1443953>
- Mauss, I. B., Levenson, R. W., McCarter, L., Wilhelm, F. H., & Gross, J. J. (2005). The tie that binds? Coherence among emotion experience, behavior, and physiology. *Emotion*, 5(2), 175–190. <https://doi.org/10.1037/1528-3542.5.2.175>
- McCall, C., Hildebrandt, L. K., Bornemann, B., & Singer, T. (2015). Physiophenomenology in retrospect: Memory reliably reflects physiological arousal during a prior threatening experience. *Consciousness and Cognition*, 38, 60–70.
<https://doi.org/10.1016/j.concog.2015.09.011>
- McHugh M. L. (2012). Interrater reliability: The kappa statistic. *Biochemia Medica*, 22(3), 276–282.
- Mehling, W. E., Acree, M., Stewart, A., Silas, J., & Jones, A. (2018). The Multidimensional Assessment of Interoceptive Awareness, Version 2 (MAIA-2). *PLOS ONE*, 13(12), e0208034. <https://doi.org/10.1371/journal.pone.0208034>

- Melcher, D., & Bacci, F. (2013). Perception of emotion in abstract artworks. In *Progress in Brain Research*, 204, 191–216). Elsevier. <https://doi.org/10.1016/B978-0-444-63287-6.00010-5>
- Menninghaus, W., Wagner, V., Hanich, J., Wassiliwizky, E., Kuehnast, M., & Jacobsen, T. (2015). Towards a psychological construct of Being moved. *PLOS ONE*, 10(6), e0128451. <https://doi.org/10.1371/journal.pone.0128451>
- Mikkelsen, M. B., O'Toole, M. S., Lyby, M. S., Wallot, S., & Mehlsen, M. (2019). Emotional reactivity and interoceptive sensitivity: Exploring the role of age. *Psychonomic Bulletin & Review*, 26(4), 1440–1448. <https://doi.org/10.3758/s13423-019-01603-y>
- Millman, L. S. M., Hunter, E. C. M., Terhune, D. B., & Orgs, G. (2023). Online structured dance/movement therapy reduces bodily detachment in depersonalization-derealization disorder. *Complementary Therapies in Clinical Practice*, 51, 101749. <https://doi.org/10.1016/j.ctcp.2023.101749>
- Murphy, J. (2023). Interoception: Where do we go from here? *Quarterly Journal of Experimental Psychology*, 174702182311727. <https://doi.org/10.1177/17470218231172725>
- Nummenmaa, L., & Hari, R. (2023). Bodily feelings and aesthetic experience of art. *Cognition and Emotion*, 0(0), 1–14. <https://doi.org/10.1080/02699931.2023.2183180>
- Palmer, C., Ainley, V., & Tsakiris, M. (2019). *Fine Tuning Your Heart: A novel method for measuring interoceptive accuracy* [Preprint]. PsyArXiv. <https://doi.org/10.31234/osf.io/qz7r9>
- Peirce, J. W. (2007). PsychoPy—Psychophysics Software in Python. *Journal of Neuroscience Methods*, 162, 8–13. <http://dx.doi.org/10.1016/j.jneumeth.2006.11.017>

- Pelowski, M., Markey, P. S., Luring, J. O., & Leder, H. (2016). Visualizing the impact of art: An update and comparison of current psychological models of art experience. *Frontiers in Human Neuroscience*, 10. <https://doi.org/10.3389/fnhum.2016.00160>
- Pelowski, M., Specker, E., Gerger, G., Leder, H., & Weingarden, L. S. (2020). Do you feel like I do? A study of spontaneous and deliberate emotion sharing and understanding between artists and perceivers of installation art. *Psychology of Aesthetics, Creativity, and the Arts*, 14(3), 276–293. <https://doi.org/10.1037/aca0000201>
- Plans, D., Ponzo, S., Morelli, D., Cairo, M., Ring, C., Keating, C. T., Cunningham, A. C., Catmur, C., Murphy, J., & Bird, G. (2021). Measuring interoception: The phase adjustment task. *Biological Psychology*, 165, 108171. <https://doi.org/10.1016/j.biopsycho.2021.108171>
- Pollatos, O., Herbert, B. M., Matthias, E., & Schandry, R. (2007). Heart rate response after emotional picture presentation is modulated by interoceptive awareness. *International Journal of Psychophysiology*, 63, 117–124. <https://doi.org/10.1016/j.ijpsycho.2006.09.003>
- Pollatos, O., Traut-Mattausch, E., Schroeder, H., & Schandry, R. (2007). Interoceptive awareness mediates the relationship between anxiety and the intensity of unpleasant feelings. *Journal of Anxiety Disorders*, 21(7), 931–943. <https://doi.org/10.1016/j.janxdis.2006.12.004>
- Pollatos, O., Ferentzi, E. (2018). *Embodiment of Emotion Regulation*. In: Hauke, G., Kritikos, A. (eds) *Embodiment in Psychotherapy*. Springer, Cham. https://doi.org/10.1007/978-3-319-92889-0_4
- Posner, J., Russell, J. A., & Peterson, B. S. (2005). The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and

psychopathology. *Development and Psychopathology*, 17(03).

<https://doi.org/10.1017/S0954579405050340>

Prinz, J. J. (2004). *Gut reactions: A perceptual theory of emotion*. Oxford: Oxford University Press.

Prinz, J. (2011). Emotion and aesthetic value. In E. Schellekens & P. Goldie (Eds.), *The aesthetic mind: Philosophy and psychology* (pp. 71–88). Oxford: Oxford University Press.

Prinz, J. (2014). Seeing with Feeling. In G. Currie, M. Kieran, A. Meskin, & J. Robson (Eds.), *Aesthetics and the Sciences of Mind* (pp. 143–158). New York: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199669639.003.0008>

Prinz, J. J. (2007). *The emotional construction of morals*. New York: Oxford University Press.

Quadt, L., Garfinkel, S. N., Mulcahy, J. S., Larsson, D. E., Silva, M., Jones, A.-M., Strauss, C., & Critchley, H. D. (2021). Interoceptive training to target anxiety in autistic adults (ADIE): A single-center, superiority randomized controlled trial. *EClinicalMedicine*, 39, 101042. <https://doi.org/10.1016/j.eclinm.2021.101042>

Revelle, W. (2017) psych: Procedures for Personality and Psychological Research, Northwestern University, Evanston, Illinois, USA, <https://CRAN.R-project.org/package=psych> Version = 1.7.8.

Ring, C., Brener, J., Knapp, K., & Mailloux, J. (2015). Effects of heartbeat feedback on beliefs about heart rate and heartbeat counting: A cautionary tale about interoceptive awareness. *Biological Psychology*, 104, 193–198.

<https://doi.org/10.1016/j.biopsycho.2014.12.010>

Rodway, P., Kirkham, J., Schepman, A., Lambert, J., & Locke, A. (2016). The development of shared liking of representational but not abstract art in primary school children and

their justifications for liking. *Frontiers in Human Neuroscience*, 10.

<https://www.frontiersin.org/articles/10.3389/fnhum.2016.00021>

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178. <https://doi.org/10.1037/h0077714>

Schandry R. (1981). Heartbeat perception and emotional experience. *Psychophysiology*, 18(4), 483–488. <https://doi.org/10.1111/j.1469-8986.1981.tb02486.x>

Schepman, A., & Rodway, P. (2021). Shared meaning in representational and abstract visual art: An empirical study. *Psychology of Aesthetics, Creativity, and the Arts*, 15(3), 458–469. <https://doi.org/10.1037/aca0000279>

Schepman, A., Rodway, P., & Pullen, S. J. (2015). Greater cross-viewer similarity of semantic associations for representational than for abstract artworks. *Journal of Vision*, 15(14), 12. <https://doi.org/10.1167/15.14.12>

Schepman, A., Rodway, P., Pullen, S. J., & Kirkham, J. (2015). Shared liking and association valence for representational art but not abstract art. *Journal of Vision*, 15(5), 11. <https://doi.org/10.1167/15.5.11>

Schindler, I., Hosoya, G., Menninghaus, W., Beermann, U., Wagner, V., Eid, M., & Scherer, K. R. (2017). Measuring aesthetic emotions: A review of the literature and a new assessment tool. *PLOS ONE*, 12(6), e0178899. <https://doi.org/10.1371/journal.pone.0178899>

Schlotz, W., Wallot, S., Omigie, D., Masucci, M. D., Hoelzmann, S. C., & Vessel, E. A. (2021). The Aesthetic Responsiveness Assessment (AReA): A screening tool to assess individual differences in responsiveness to art in English and German. *Psychology of Aesthetics, Creativity, and the Arts*, 15(4), 682–696. <https://doi.org/10.1037/aca0000348>

- Schuetz, S. A., Zucker, N. L., & Smoski, M. J. (2021). Do interoceptive accuracy and interoceptive sensibility predict emotion regulation? *Psychological Research*, 85(5), 1894–1908. <https://doi.org/10.1007/s00426-020-01369-2>
- Silvia, P. J., Fayn, K., Nusbaum, E. C., & Beaty, R. E. (2015). Openness to experience and awe in response to nature and music: Personality and profound aesthetic experiences. *Psychology of Aesthetics, Creativity, and the Arts*, 9(4), 376–384. <https://doi.org/10.1037/aca0000028>
- Silvia, P. J., & Nusbaum, E. C. (2011). On personality and piloerection: Individual differences in aesthetic chills and other unusual aesthetic experiences. *Psychology of Aesthetics, Creativity, and the Arts*, 5(3), 208–214. <https://doi.org/10.1037/a0021914>
- Specker, E., Forster, M., Brinkmann, H., Boddy, J., Pelowski, M., Rosenberg, R., & Leder, H. (2020). The Vienna Art Interest and Art Knowledge Questionnaire (VAIAK): A unified and validated measure of art interest and art knowledge. *Psychology of Aesthetics, Creativity, and the Arts*, 14(2), 172–185. <https://doi.org/10.1037/aca0000205>
- Specker, E., Tinio, P. P. L., & Van Elk, M. (2017). Do you see what I see? An investigation of the aesthetic experience in the laboratory and museum. *Psychology of Aesthetics, Creativity, and the Arts*, 11(3), 265–275. <https://doi.org/10.1037/aca0000107>
- Suksasilp, C., & Garfinkel, S. N. (2022). Towards a comprehensive assessment of interoception in a multi-dimensional framework. *Biological Psychology*, 168, 108262. <https://doi.org/10.1016/j.biopsycho.2022.108262>
- Tarvainen, M. P., Niskanen, J.-P., Lipponen, J. A., Ranta-aho, P. O., & Karjalainen, P. A. (2014). Kubios HRV – Heart rate variability analysis software. *Computer Methods and Programs in Biomedicine*, 113(1), 210–220. <https://doi.org/10.1016/j.cmpb.2013.07.024>

- Taylor, G. J., Michael Bagby, R., & Parker, J. D. A. (1991). The alexithymia construct: A potential paradigm for psychosomatic medicine. *Psychosomatics*, 32(2), 153–164.
[https://doi.org/10.1016/S0033-3182\(91\)72086-0](https://doi.org/10.1016/S0033-3182(91)72086-0)
- Tinio, P., & Specker, E. (2023). For Emotion’s Sake ... The Centrality of Emotions in the Art Experience. In Z. Ivcevic, J. Hoffmann, & J. Kaufman (Eds.), *The Cambridge Handbook of Creativity and Emotions* (Cambridge Handbooks in Psychology, pp. 358-375). Cambridge: Cambridge University Press. doi:10.1017/9781009031240.024
- Uraguchi, M., Maulina, V. V. R., & Ohira, H. (2022). Interoceptive accuracy correlates with precision of time perception in the millisecond range. *Frontiers in Neuroscience*, 16, 993491. <https://doi.org/10.3389/fnins.2022.993491>
- van Paasschen, J., Bacci, F., & Melcher, D. P. (2015). The influence of art expertise and training on emotion and preference ratings for representational and abstract artworks. *PLoS ONE*, 10(8), Article e0134241. <https://doi.org/10.1371/journal.pone.0134241>
- Vartanian, O. (2018). *Internal orientation in aesthetic experience* (K. Christoff & K. C. R. Fox, Eds.; Vol. 1). Oxford, UK: Oxford University Press.
<https://doi.org/10.1093/oxfordhb/9780190464745.013.17>
- Vartanian, O., & Skov, M. (2014). Neural correlates of viewing paintings: Evidence from a quantitative meta-analysis of functional magnetic resonance imaging data. *Brain and Cognition*, 87, 52–56. <https://doi.org/10.1016/j.bandc.2014.03.004>
- Ventura-Bort, C., Wendt, J., & Weymar, M. (2021). The role of interoceptive sensibility and emotional conceptualization for the experience of emotions. *Frontiers in Psychology*, 12, 712418. <https://doi.org/10.3389/fpsyg.2021.712418>
- Vessel, E. A., & Rubin, N. (2010). Beauty and the beholder: Highly individual taste for abstract, but not real-world images. *Journal of vision*, 10(2), 1–14.
<https://doi.org/10.1167/10.2.18>

- Vessel, E. A., Starr, G. G., & Rubin, N. (2012). The brain on art: Intense aesthetic experience activates the default mode network. *Frontiers in Human Neuroscience*, 6.
<https://doi.org/10.3389/fnhum.2012.00066>
- Vessel, E. A., Starr, G. G., & Rubin, N. (2013). Art reaches within: Aesthetic experience, the self and the default mode network. *Frontiers in Neuroscience*, 7.
<https://doi.org/10.3389/fnins.2013.00258>
- Ware, M., O'Connor, P., Bub, K., Backus, D., & McCully, K. (2023). Investigating relationships among interoceptive awareness, emotional susceptibility, and fatigue in persons with multiple sclerosis. *International Journal of MS Care*, 25(2), 75–81.
<https://doi.org/10.7224/1537-2073.2022-007>
- Wiens, S., Mezzacappa, E. S., & Katkin, E. S. (2000). Heartbeat detection and the experience of emotions. *Cognition & Emotion*, 14(3), 417–427.
<https://doi.org/10.1080/026999300378905>
- Williams, P. G., Johnson, K. T., Curtis, B. J., King, J. B., & Anderson, J. S. (2018). Individual differences in aesthetic engagement are reflected in resting-state fMRI connectivity: Implications for stress resilience. *NeuroImage*, 179, 156–165.
<https://doi.org/10.1016/j.neuroimage.2018.06.042>
- Young, K. A., Gandevia, S. C., & Giummarra, M. J. (2017). Vicarious pain responders and emotion: Evidence for distress rather than mimicry. *Psychophysiology*, 54(7), 1081–1095. <https://doi.org/10.1111/psyp.12865>
- Zaki, J., Davis, J. I., & Ochsner, K. N. (2012). Overlapping activity in anterior insula during interoception and emotional experience. *NeuroImage*, 62(1), 493–499.
<https://doi.org/10.1016/j.neuroimage.2012.05.012>

Zamariola, G., Frost, N., Van Oost, A., Corneille, O., & Luminet, O. (2019). Relationship between interoception and emotion regulation: New evidence from mixed methods.

Journal of Affective Disorders, 246, 480–485. <https://doi.org/10.1016/j.jad.2018.12.101>

Figure A1. All 60 artworks plotted along mean ratings on *Valence/Arousal* collected in Study 2 for mean *Being Moved* and *Liking* ratings per artwork. Dashed lines represent the means of *Arousal/Valence* ratings. See list of artworks in Supplementary Material (Table S1).

